

Modelli e Algoritmi di Ottimizzazione

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- a) mathematical modeling for decision-making and simulation
- b) applications to real-life decision problems
- c) solve the problem using GAMS
- d) analyse the solution obtained

Operations research, also known as **management science**,

deals with mathematical models and solution algorithms

for analysing and solving real-life decision-making problems.

Examples from the energy sector:

- day-ahead market clearing: power dispatching and electricity prices
- optimizing power generation for self-consumption and electricity purchase from the grid
- active distribution networks

The real-life problem is represented by a mathematical model in which

- the decisions to be taken are represented by variables
- each **decision variable** takes values in a specific set, e.g. $\mathbb{R}$, $\mathbb{R}_+$, $\mathbb{N}$, $\{0, 1\}$
- **functional constraints** express the interdependency among decision variables:
these constraints in general define infinitely many alternative decisions (**feasible region**)
- an objective function defines the selection criterion among the infinitely many feasible alternative solutions, by which an **optimal feasible solution** is determined

Example: developing the model of a 2-variable production scheduling problem

We have to schedule the production of a plant over a 3-month planning period.

The planning period is discretized in weeks (13 weeks in 3-month planning period).

The plant is used to manufacture two products, P1 and P2, with the following time requirements:

- P1: $1 \left[\frac{\text{hour}}{\text{batch}} \right]$
- P2: $2 \left[\frac{\text{hour}}{\text{batch}} \right]$

Each week the available plant capacity is $10 \left[\frac{\text{hours}}{\text{week}} \right]$.

According to a market research, the maximum quantities that can be sold each week are

- P1: at most $4 \left[\frac{\text{batch}}{\text{week}} \right]$
- P2: any quantity.

Unit profits

- P1: $2 \left[\frac{\text{k€}}{\text{batch}} \right]$
- P2: $3 \left[\frac{\text{k€}}{\text{batch}} \right]$

Represent the decision problem by a mathematical model

To be decided:

the weekly production rates $\left[\frac{\text{batch}}{\text{week}}\right]$ of P1 and P2, denoted by x_1 and x_2 respectively

x_1 and x_2 are called **decision variables**

x_1 and x_2 must be constrained to take on real nonnegative values

- negative values are meaningless
- fractional values are not meaningless, since x_1 and x_2 express weekly production rates

Non-negativity constraints (to be expressed explicitly)

$$x_1 \geq 0, \quad x_2 \geq 0$$

Conditions to be met:

- in each week the total working time cannot exceed $10 \left[\frac{\text{hour}}{\text{week}} \right]$
- weekly production rate of P1 cannot exceed $4 \left[\frac{\text{batch}}{\text{week}} \right]$

The total working time cannot exceed $10 \left[\frac{\text{hours}}{\text{week}} \right]$

- If the **Proportionality hypothesis** holds, for each product the weekly working time is proportional to the quantity produced per week

$$1 \cdot x_1 \left[\frac{\text{hours}}{\text{batch}} \right] \cdot \left[\frac{\text{batch}}{\text{week}} \right]$$

$$2 \cdot x_2 \left[\frac{\text{hours}}{\text{batch}} \right] \cdot \left[\frac{\text{batch}}{\text{week}} \right]$$

- If the **Additivity hypothesis** holds, the total weekly working time for all products is the sum

$$x_1 + 2 x_2 \leq 10 \left[\frac{\text{hours}}{\text{week}} \right]$$

Weekly production rate of P1 cannot exceed the weekly market demand

$$x_1 \leq 4 \quad \left[\frac{batch}{week} \right]$$

upper bound constraint

Preferred solution: find the weekly production rates that maximize the weekly profit

- **Proportionality:** for both products, weekly profit is proportional to the amount produced

$$2 \cdot x_1 \left[\frac{k\text{€}}{\text{batch}} \right] \cdot \left[\frac{\text{batch}}{\text{week}} \right]$$

$$3 \cdot x_2 \left[\frac{k\text{€}}{\text{batch}} \right] \cdot \left[\frac{\text{batch}}{\text{week}} \right]$$

- **Additivity:** total weekly profit, represented by **objective variable** ω

$$\omega = 2 x_1 + 3 x_2 \left[\frac{k\text{€}}{\text{week}} \right] \quad \textbf{objective function}$$

maximize ω

ω is a FREE (i.e. unconstrained) variable

Mathematical model

$$\begin{array}{ll} \max \omega = 2x_1 + 3x_2 & \text{objective function} \\ \left. \begin{array}{l} x_1 + 2x_2 \leq 10 \\ x_1 \leq 4 \end{array} \right\} & \text{functional constraints} \\ x_1 \geq 0, x_2 \geq 0 & \text{non-negativity constraints} \end{array}$$

Mathematical model

$$\max \omega = 2x_1 + 3x_2$$

objective function

$$x_1 + 2x_2 \leq 10$$

$$x_1 \leq 4$$

functional constraints

$$x_1 \geq 0, x_2 \geq 0$$

non-negativity constraints

feasible region

Mathematical model

$$\max \omega = 2x_1 + 3x_2$$

objective function

$$x_1 + 2x_2 \leq 10$$

$$x_1 \leq 4$$

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$$x_1 \geq 0, x_2 \geq 0$$

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feasible region

Linear Programming (LP) model

- all decision variables $x \in \mathbb{R}$
- objective function and constraint functions are linear functions of the decision variables

Summarizing, to solve the decision problem we have to determine a **solution** that

A. satisfies a set of conditions

- the conditions are expressed in mathematical form as equations and inequalities
- a solution that satisfies all the requested conditions is called a **feasible solution**
- the set of all solutions that satisfy the requested conditions is called **feasible region**
(from which the decision maker must select a solution to the decision problem)

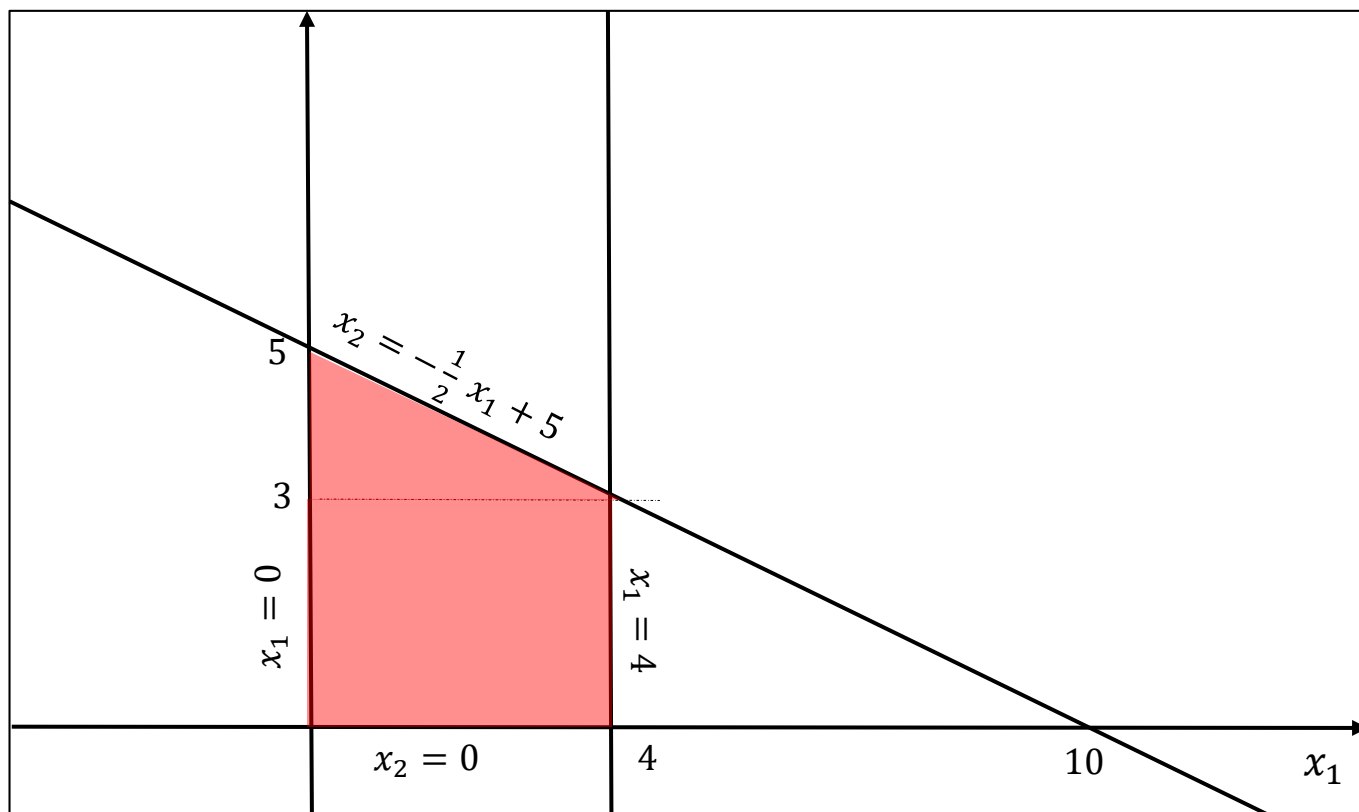
B. optimizes an evaluation criterion defining a *merit order* within the solution set

Graphical representation of the feasible region

$$x_1 + 2x_2 \leq 10$$

$$x_1 \leq 4$$

$$x_1 \geq 0 \quad x_2 \geq 0$$



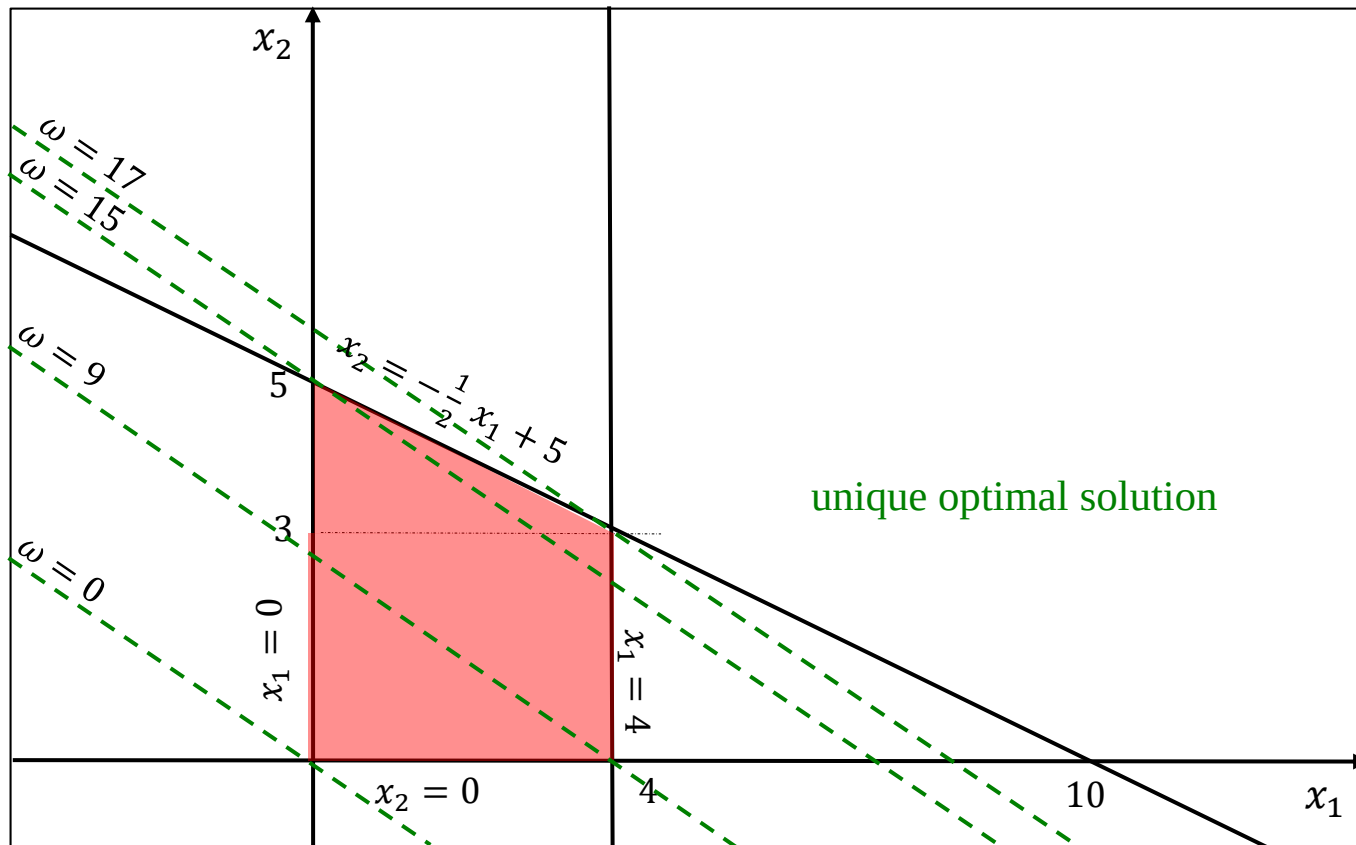
$$\omega = 2x_1 + 3x_2 \quad x_2 = -\frac{2}{3}x_1 + \left(\frac{\omega}{3}\right)$$

The intercept q is related to the objective variable ω

This mathematical expression represents a bundle of parallel straight lines with slope $-\frac{2}{3}$

$\max \omega$

Since $q = \frac{\omega}{3}$, ω is maximized if q is maximized



unique optimal solution:

Vertex B

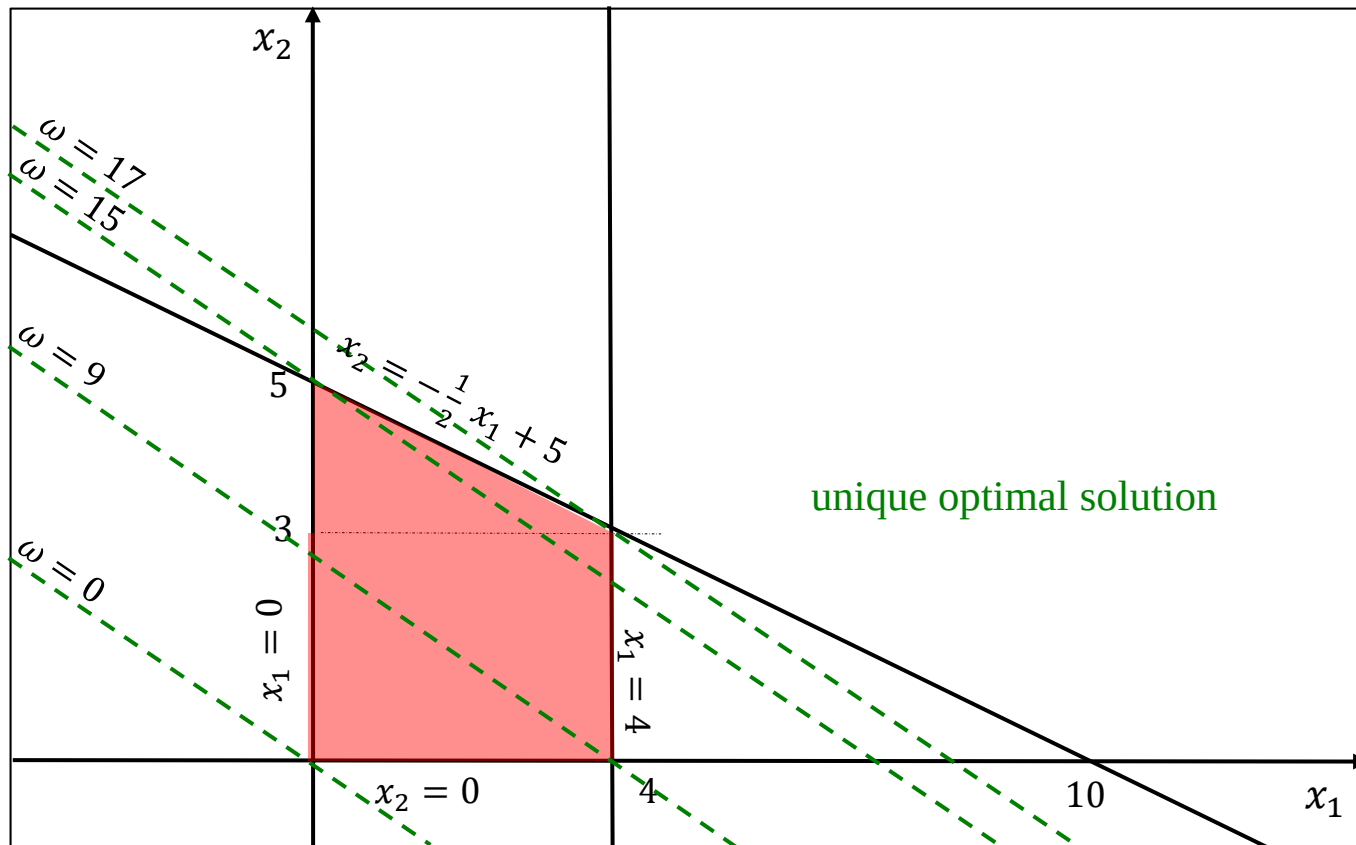
weekly production rate of P1: $x_1 = 4$

weekly production rate of P2: $x_2 = 3$

weekly profit: $\omega = 17$

at point B two constraint boundaries intersect:
functional constraints are satisfied as equalities

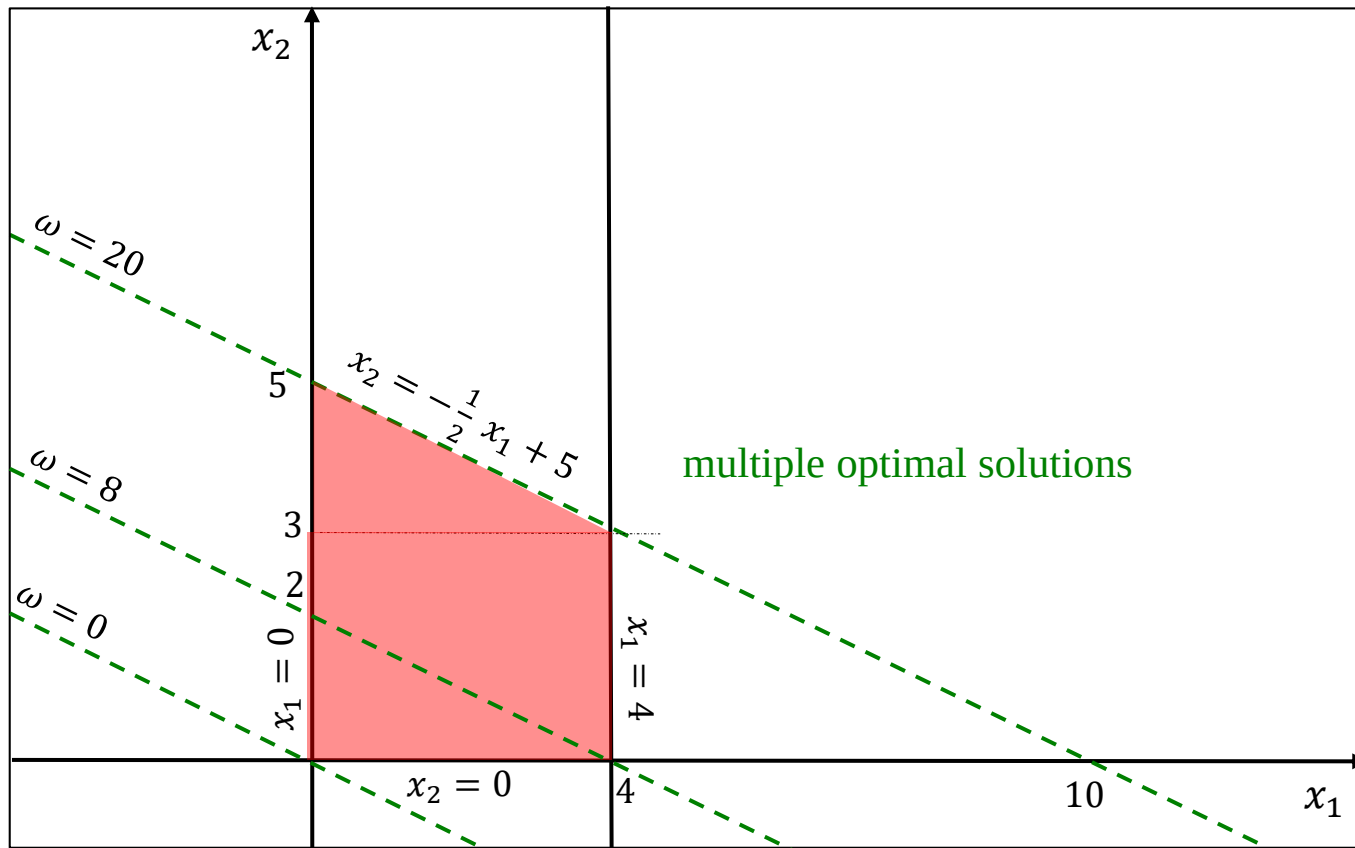
$$\text{B:} \begin{cases} x_1 + 2x_2 = 10 \\ x_1 = 4 \end{cases}$$



$$\omega = 2x_1 + 4x_2 \quad \left| \quad x_2 = -\frac{2}{4}x_1 + \left(\frac{\omega}{4}\right) \quad \text{The intercept } q \text{ is related to the objective variable } \omega\right.$$

This mathematical expression represents a bundle of parallel straight lines with slope $-\frac{1}{2}$

$\max \omega$ Since $q = \frac{\omega}{4}$, ω is maximized if q is maximized



Problem 2: design of a radiotherapy treatment

In radiotherapy treatment, beams of ionizing radiation are sent to the tumor mass.

Such radiation, however,

is also released on tissues outside the beam path (scattering)
and damages both cancerous and healthy tissues.

To modulate the **intensity of radiation** to be used in treatment,
an analysis of radiation absorption by tissues is needed.

Consider the case in which the treatment requires the use of two beams:

the results of the radiation absorption analysis are summarized in the following table

Average absorption, in <i>kilorads</i> , per <i>kilorad</i> of radiation intensity at the point of entry		
	beam 1	beam 2
healthy tissues	0.4	0.5
critical tissues	0.3	0.1
tumor core	0.6	0.4
tumor region	0.5	0.5

The absorption of radiation from the two beams is **additive**

Absorption of different areas:

- healthy tissues: as little as possible
- critical tissues: at most 2.7 kilorad
- tumor core: at least 6 kilorad
- tumor region: exactly 6 kilorad

x_1 [*kilorad*]: intensity of beam 1 at entry point

x_2 [*kilorad*]: Intensity of beam 2 at entry point

absorption [<i>kilorad</i>]	
healthy tissues	$0.4 x_1 + 0.5 x_2$
critical tissues	$0.3 x_1 + 0.1 x_2$
tumor core	$0.6 x_1 + 0.4 x_2$
tumor region	$0.5 x_1 + 0.5 x_2$

Mathematical model

$$\min \varphi = 0.4 x_1 + 0.5 x_2$$

minimize the radiation absorbed by the healthy tissues

$$\text{s.t. } 0.3 x_1 + 0.1 x_2 \leq 2.7$$

upper bound to radiation absorbed by the critical tissues

$$0.6 x_1 + 0.4 x_2 \geq 6.0$$

lower bound to the radiation absorbed by the tumor core

$$0.5 x_1 + 0.5 x_2 = 6.0$$

absorbed radiation required in the tumor region

$$x_1 \geq 0, x_2 \geq 0$$

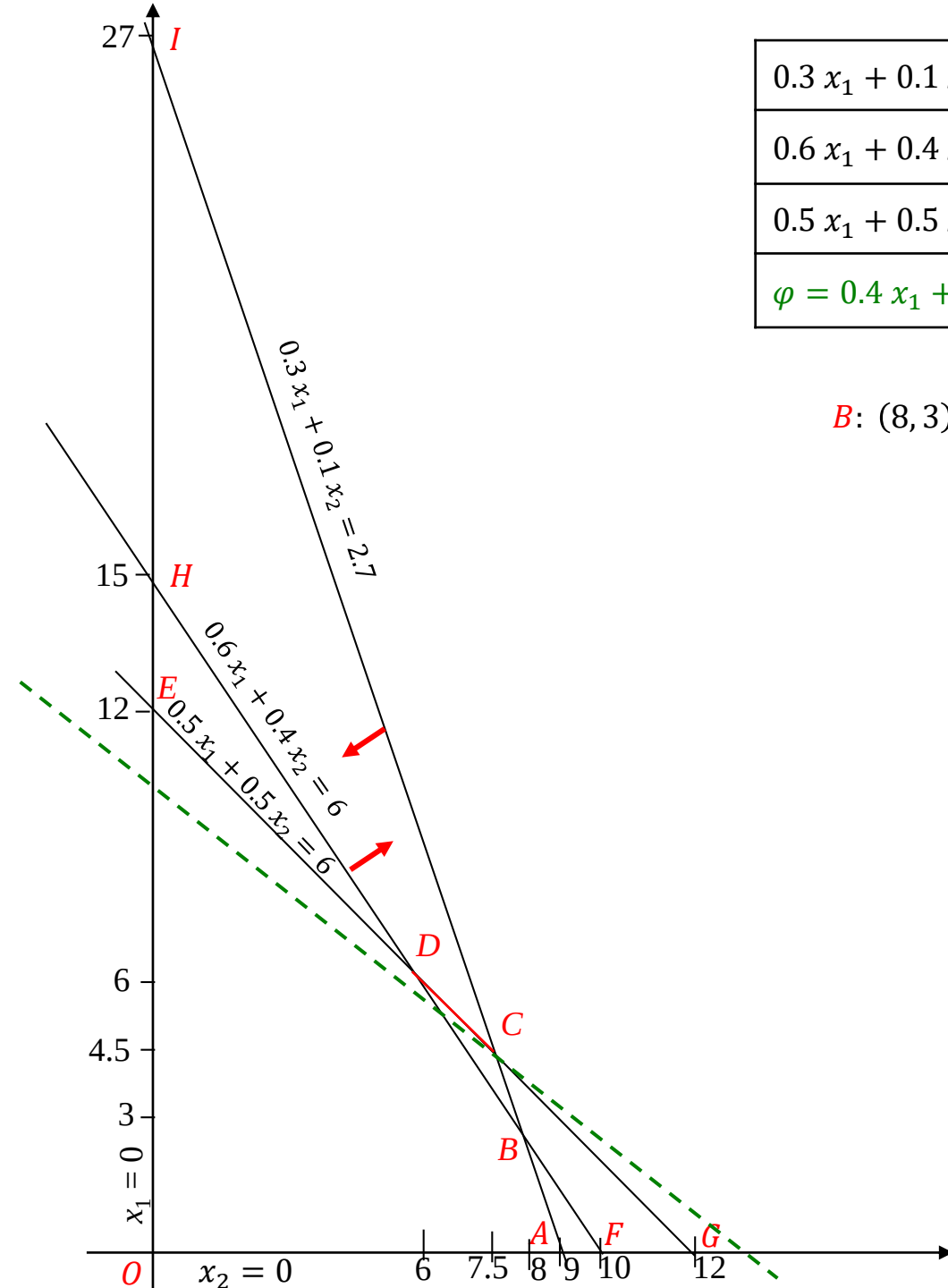
the intensities of rays 1 and 2 at their respective entry points are expressed by nonnegative real numbers

$0.3 x_1 + 0.1 x_2 \leq 2.7$	$\Leftrightarrow$	$x_2 \leq -3 x_1 + 27$
$0.6 x_1 + 0.4 x_2 \geq 6.0$	$\Leftrightarrow$	$x_2 \geq -\frac{3}{2} x_1 + 15$
$0.5 x_1 + 0.5 x_2 = 6.0$	$\Leftrightarrow$	$x_2 = -x_1 + 12$
$\varphi = 0.4 x_1 + 0.5 x_2$	$\Leftrightarrow$	$x_2 = -\frac{4}{5} x_1 + 2 \varphi$

$B: (8, 3)$

$C: (7.5, 4.5)$

$D: (6, 6)$



NOTE: If the absorbed radiation in the tumor region is to be

"exactly 5.5 kilorad "

instead of "exactly 6.0 kilorad ", the third functional constraint becomes

$$0.5 x_1 + 0.5 x_2 = 5.5 \quad \Leftrightarrow \quad x_2 = -x_1 + 11$$

The feasible region changes as shown in the graph

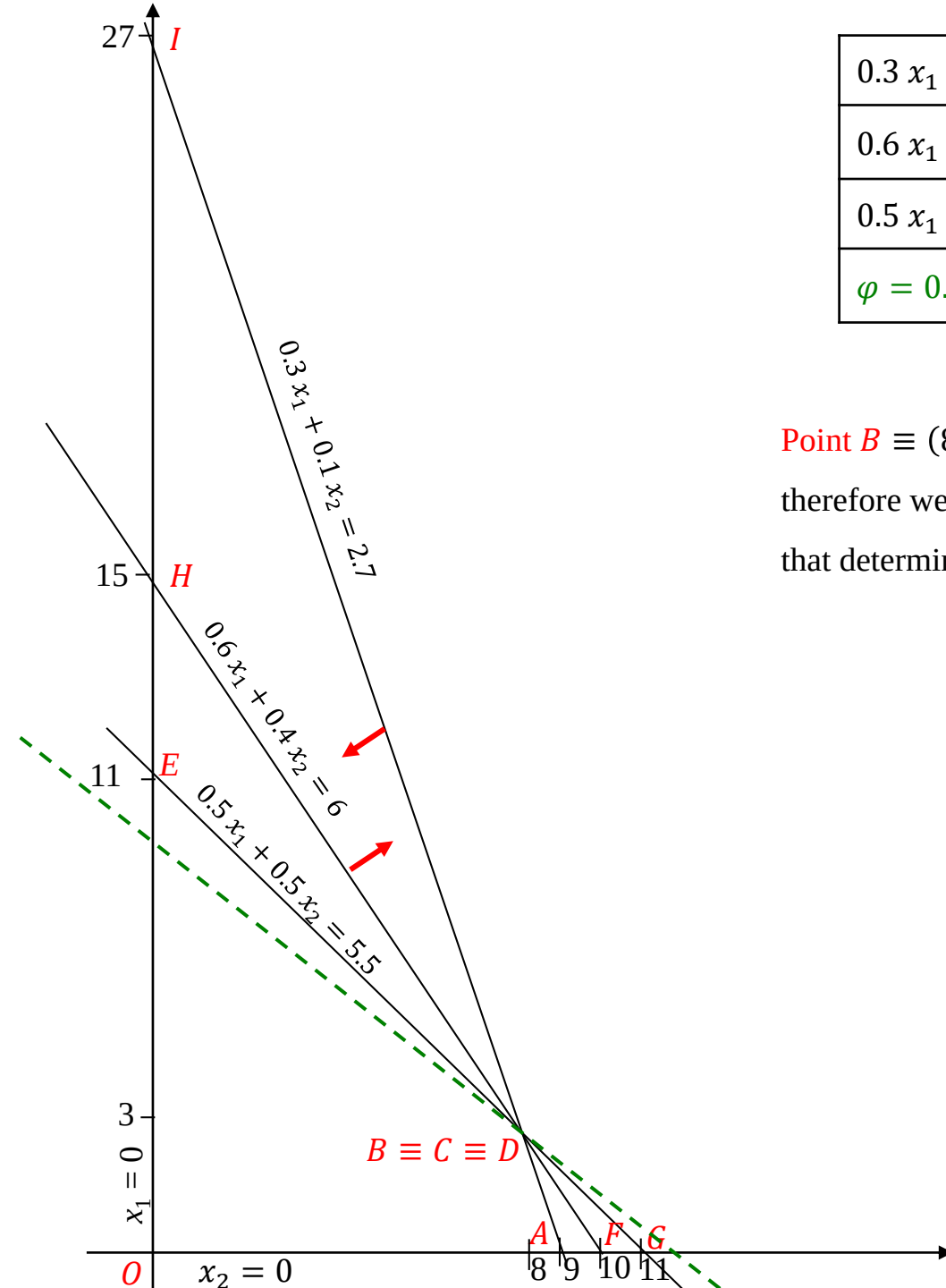
$0.3 x_1 + 0.1 x_2 \leq 2.7$	$\Leftrightarrow$	$x_2 \leq -3 x_1 + 27$
$0.6 x_1 + 0.4 x_2 \geq 6.0$	$\Leftrightarrow$	$x_2 \geq -\frac{3}{2} x_1 + 15$
$0.5 x_1 + 0.5 x_2 = 5.5$	$\Leftrightarrow$	$x_2 = -x_1 + 11$
$\varphi = 0.4 x_1 + 0.5 x_2$	$\Leftrightarrow$	$x_2 = -\frac{4}{5} x_1 + 2 \varphi$

Point $B \equiv (8, 3)$ is the intersection of 3 constraint boundaries, therefore we may define 3 distinct linear systems that determine such point, namely

$$\begin{cases} 0.3 x_1 + 0.1 x_2 = 2.7 \\ 0.6 x_1 + 0.4 x_2 = 6.0 \end{cases}$$

$$\begin{cases} 0.3 x_1 + 0.1 x_2 = 2.7 \\ 0.5 x_1 + 0.5 x_2 = 5.5 \end{cases}$$

$$\begin{cases} 0.6 x_1 + 0.4 x_2 = 6.0 \\ 0.5 x_1 + 0.5 x_2 = 5.5 \end{cases}$$



NOTE: If the absorbed radiation in the tumor region is to be

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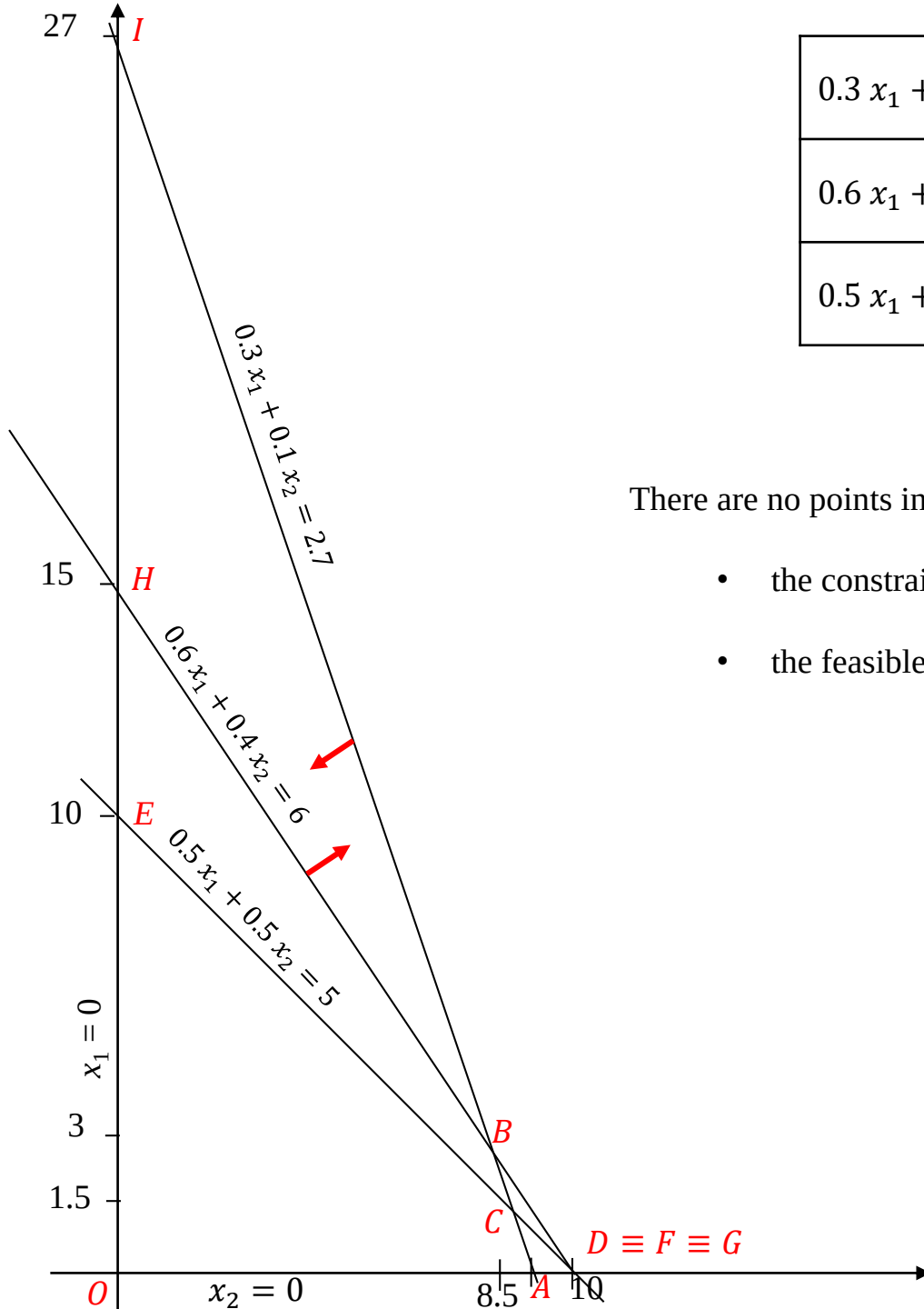
$$0.5 x_1 + 0.5 x_2 = 5 \quad \Leftrightarrow \quad x_2 = -x_1 + 10$$

The feasible region changes as shown in the graph

$0.3 x_1 + 0.1 x_2 \leq 2.7$	$\Leftrightarrow$	$x_2 \leq -3 x_1 + 27$
$0.6 x_1 + 0.4 x_2 \geq 6.0$	$\Leftrightarrow$	$x_2 \geq -\frac{3}{2} x_1 + 15$
$0.5 x_1 + 0.5 x_2 = 5.0$	$\Leftrightarrow$	$x_2 = -x_1 + 10$

There are no points in the plane (x_1, x_2) that satisfy all the constraints:

- the constraints are mutually incompatible
- the feasible region is empty.



Standardizing the LP model

In problem

$$\begin{array}{ll}\max \omega = & 2 x_1 + 3 x_2 \\ \text{s.t.} & x_1 + 2 x_2 \leq 10 \\ & x_1 \leq 4 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

the feasible region and the objective function are reformulated as follows:

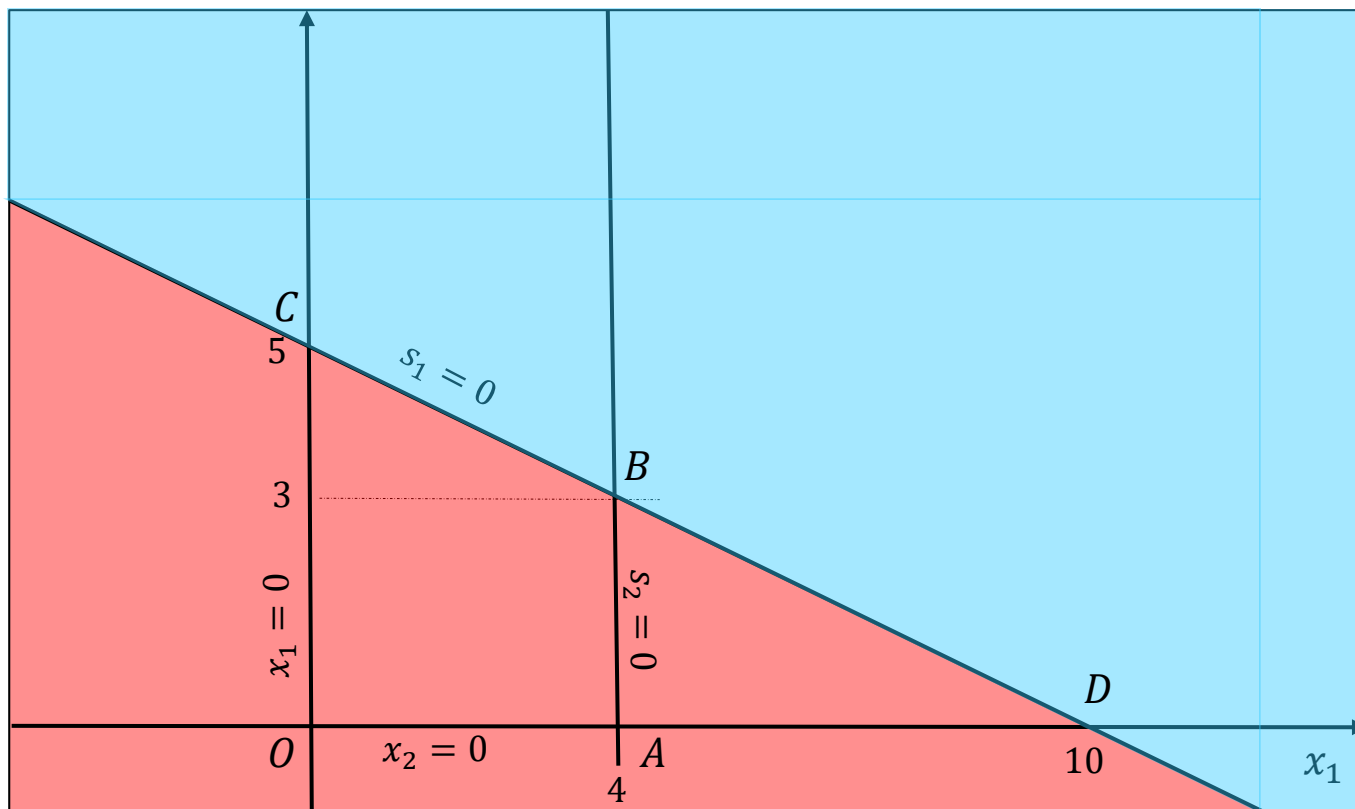
(a) reformulation of the feasible region

$$\begin{array}{ll} x_1 + 2 x_2 \leq 10 \\ x_1 \leq 4 \\ x_1 \geq 0, x_2 \geq 0 \end{array} \quad \rightarrow \quad \begin{array}{llll} x_1 + 2 x_2 + s_1 & = & 10 \\ x_1 + s_2 & = & 4 \\ x_1 \geq 0 & x_2 \geq 0 & s_1 \geq 0 & s_2 \geq 0 \end{array}$$

$$x_1 + 2x_2 \leq 10$$

$$x_1 + 2x_2 + s_1 = 10$$

$$s_1 \geq 0$$

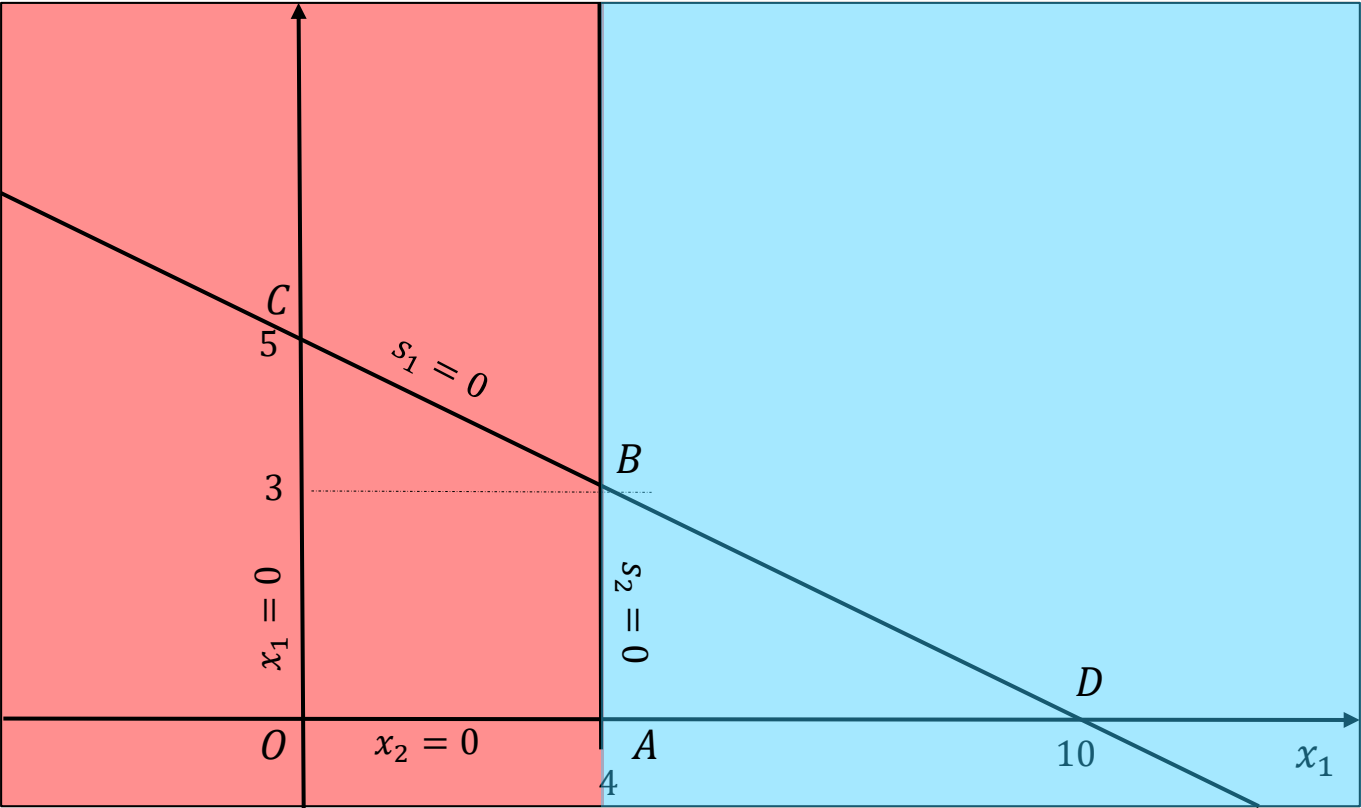


$s_1 = 0$	$x_1 + 2x_2 = 10$
$s_1 > 0$	$x_1 + 2x_2 < 10$
$s_1 < 0$	$x_1 + 2x_2 > 10$

$$x_1 \leq 4$$

$$x_1 + s_2 = 4$$

$$s_2 \geq 0$$



$s_2 = 0$	$x_1 = 4$
$s_2 > 0$	$x_1 < 4$
$s_2 < 0$	$x_1 > 4$

Alternative representation of the LP problem

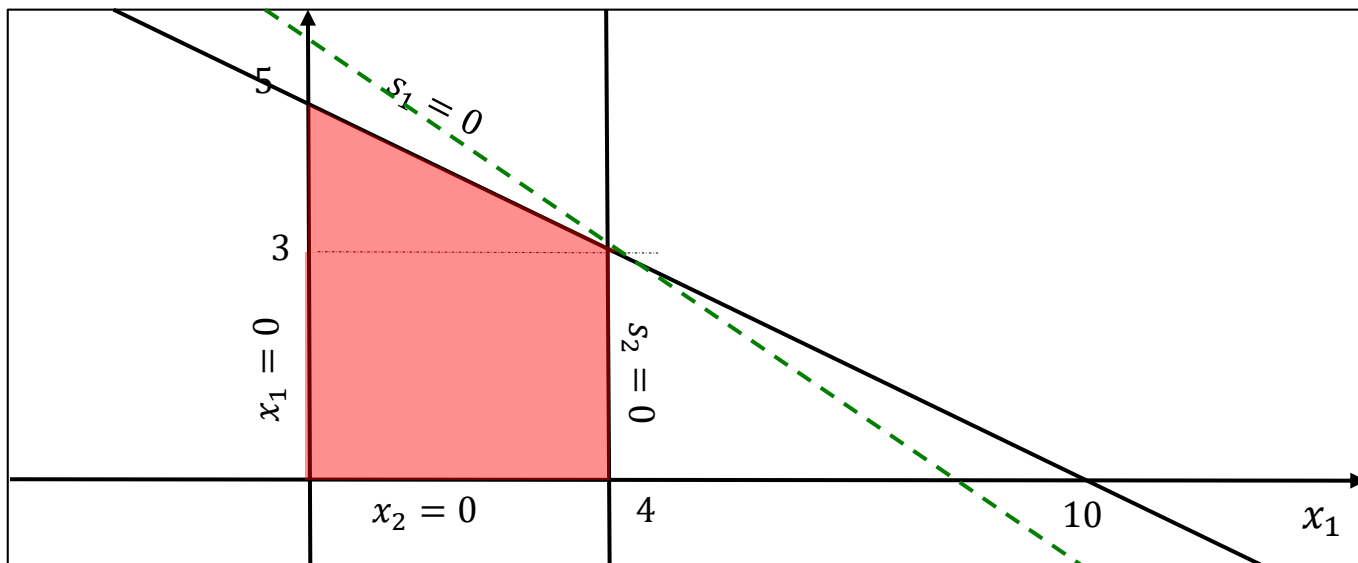
$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + s_2 = 4$$

$$-2x_1 - 3x_2 + \omega = 0$$

$$x_1 \geq 0 \quad x_2 \geq 0 \quad s_1 \geq 0 \quad s_2 \geq 0 \quad \max \omega$$

- 2 linear equations in the 4 unknowns x_1, x_2, s_1, s_2 bounded to be nonnegative (the feasible region is the intersection of the 4 half-planes)
- linear equation that defines the objective variable ω as a function of variables x_1 and x_2 :
 ω to be maximized



Given any vector $(\hat{x}_1, \hat{x}_2, \hat{s}_1, \hat{s}_2, \hat{w})$ such that

$$\begin{aligned}\hat{x}_1 + 2\hat{x}_2 + \hat{s}_1 &= 10 \\ \hat{x}_1 + \hat{s}_2 &= 4 \\ -2\hat{x}_1 - 3\hat{x}_2 + \hat{w} &= 0\end{aligned}$$

- the pair $(\hat{x}_1, \hat{x}_2) \in \mathbb{R}^2$ identifies a decision in the space $\mathbb{R}^2$ of decisions
- the value of each slack variable $\hat{s}_i, i = 1, 2$,
indicates the position of $(\hat{x}_1, \hat{x}_2)$ with respect to the boundary of constraint i ,
i.e. whether decision $(\hat{x}_1, \hat{x}_2)$ is feasible with respect to constraint i
- $\hat{w}$ is the value of the objective function corresponding to decision $(\hat{x}_1, \hat{x}_2)$

Standardizing the LP model

$$\min \varphi = 0.4 x_1 + 0.5 x_2$$

$$0.3 x_1 + 0.1 x_2 + s_1 = 2.7$$

$$0.6 x_1 + 0.4 x_2 - \bar{s}_2 = 6$$

$$0.5 x_1 + 0.5 x_2 + t_3^+ = 6$$

$$0.5 x_1 + 0.5 x_2 - t_3^- = 6$$

$$x_1, x_2, s_1, \bar{s}_2, t_3^+, t_3^- \geq 0$$

$$\mathbf{F} = \begin{vmatrix} 0.3 & 0.1 & 1 & 0 & 0 & 0 \\ 0.6 & 0.4 & 0 & -1 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 & 1 & 0 \\ 0.5 & 0.5 & 0 & 0 & 0 & -1 \end{vmatrix}$$

$$\mathbf{w} = \begin{vmatrix} x_1 \\ x_2 \\ s_1 \\ \bar{s}_2 \\ t_3^+ \\ t_3^- \end{vmatrix}$$

$$\mathbf{b} = \begin{vmatrix} 2.7 \\ 6 \\ 6 \\ 6 \end{vmatrix}$$

$$\mathbf{c}_w^T = \begin{vmatrix} 0.4 & 0.5 & 0 & 0 & 0 & 0 \end{vmatrix}$$

$$\min \varphi = \mathbf{c}_w^T \mathbf{w} + \gamma$$

$$\text{s.t.} \quad \mathbf{F} \mathbf{w} = \mathbf{b}$$

$$\mathbf{w} \geq 0$$

$$\mathbf{F} \in \mathbb{R}^{m,n+m}, \text{rank}(\mathbf{F}) = m$$

m : number of functional constraints

n : number of decision variables

Canonical form of the underdetermined linear system $\mathbf{F} \mathbf{w} = \mathbf{b}$ with respect to a basis $\mathbf{B}$ of $\mathbf{F}$

Definitions of **basis** $\mathbf{B}$ of a coefficient matrix $\mathbf{F}$, with $\mathbf{F} \in \mathbb{R}^{m,n+m}$ and $\text{rank}(\mathbf{F}) = m$:

a basis $\mathbf{B}$ of $\mathbf{F}$ is a nonsingular square matrix of order m
consisting of m linearly independent columns of $\mathbf{F}$

Suppose the columns of $\mathbf{B}$ are the first m columns of $\mathbf{F}$ (*):

we can express $\mathbf{F}$ in the following partitioned form

$$\mathbf{F} = [\mathbf{B} \quad \mathbf{N}]$$

where

$\mathbf{B}$ is the matrix of basic columns

$\mathbf{N}$ is the matrix of non-basic columns

(*) there is no loss of generality, since any problem can be expressed in this form by suitably reordering the columns of matrix $\mathbf{F}$ and of the components of vectors $\mathbf{w}$ and $\mathbf{c}_w$

Analogously, vector $\mathbf{w}$ can be expressed in partitioned form

$$\mathbf{w} = \begin{bmatrix} \mathbf{w}_B \\ \mathbf{w}_N \end{bmatrix}$$

where

$\mathbf{w}_B$: vector of **basic variables**, i.e. variables associated to the columns of $\mathbf{B}$

$\mathbf{w}_N$: vector of **non-basic variables**, i.e. variables associated to the columns of $\mathbf{N}$

and vector $\mathbf{c}_w$ can be expressed in partitioned form

$$\mathbf{c}_w = \begin{bmatrix} \mathbf{c}_B \\ \mathbf{c}_N \end{bmatrix}$$

where

$\mathbf{c}_B$: coefficients of the basic variables $\mathbf{w}_B$ in the objective function

$\mathbf{c}_N$: coefficients of the non-basic variables $\mathbf{w}_N$ in the objective function

The underdetermined linear system

$$\mathbf{F} \mathbf{w} = \mathbf{b}$$

can be expressed as

$$\mathbf{F} \mathbf{w} = [\mathbf{B} \quad \mathbf{N}] \begin{bmatrix} \mathbf{w}_B \\ \mathbf{w}_N \end{bmatrix},$$

and equivalently

$$\mathbf{B} \mathbf{w}_B + \mathbf{N} \mathbf{w}_N = \mathbf{b}$$

By premultiplying by $\mathbf{B}^{-1}$ both sides of $\mathbf{B} \mathbf{w}_B + \mathbf{N} \mathbf{w}_N = \mathbf{b}$,

we get the linear system

$$\mathbf{w}_B + \mathbf{B}^{-1} \mathbf{N} \mathbf{w}_N = \mathbf{B}^{-1} \mathbf{b}$$

which is equivalent to the linear system $\mathbf{F} \mathbf{w} = \mathbf{b}$

and is called **canonical form associated to basis $\mathbf{B}$** .

In the canonical form associated to basis B ,

$$\mathbf{w}_B + B^{-1}N \mathbf{w}_N = B^{-1}\mathbf{b}$$

each component of $\mathbf{w}_B$ (i.e. each **basic variable**) is expressed
only in terms of non-basic variables

The set of all solutions of $F \mathbf{w} = \mathbf{b}$ is obtained by assigning
all possible values to the nonbasic variables, i.e. $\mathbf{w}_N \in \mathbb{R}^n$.

It is also useful to reformulate the objective function so that
the **objective variable is expressed only in terms of non-basic variables**.

To this aim, supposing that

- we are considering a minimization problem
- the basic variables are the first m variables in $\mathbf{w}$,

the objective function $\varphi = \mathbf{c}_w^T \mathbf{w} + \gamma$ (with φ to be minimized)

can be written as $\varphi = \mathbf{c}_B^T \mathbf{w}_B + \mathbf{c}_N^T \mathbf{w}_N + \gamma$

From the canonical form associated to basis $\mathbf{B}$ the following expression for $\mathbf{w}_B$ holds

$$\mathbf{w}_B = \mathbf{B}^{-1} \mathbf{b} - \mathbf{B}^{-1} \mathbf{N} \mathbf{w}_N$$

which we substitute in the expression of φ

$$\begin{aligned} \varphi &= \mathbf{c}_B^T (\mathbf{B}^{-1} \mathbf{b} - \mathbf{B}^{-1} \mathbf{N} \mathbf{w}_N) + \mathbf{c}_N^T \mathbf{w}_N + \gamma = \\ &= \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b} + \gamma - \underbrace{(\mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{N} - \mathbf{c}_N^T)}_{\mathbf{r}_N^T} \mathbf{w}_N = \\ &= \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b} + \gamma - \mathbf{r}_N^T \mathbf{w}_N \end{aligned}$$

$\mathbf{r}_N$ is the vector of the **relative costs of the non-basic variables**.

Basic solution of $\mathbf{F} \mathbf{w} = \mathbf{b}$ associated to basis $\mathbf{B}$

The basic solution associated to basis $\mathbf{B}$ is the solution where all non-basic variables take the 0 value:

$$\mathbf{w}_N = \mathbf{0}$$

From the general expressions $\mathbf{w}_B = \mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{N} \mathbf{w}_N$ and $\varphi = \mathbf{c}_B^T \mathbf{B}^{-1}\mathbf{b} + \gamma - \mathbf{r}_N^T \mathbf{w}_N$ it follows that

- the basic solution associated to basis $\mathbf{B}$ is

$$\mathbf{w} = \begin{bmatrix} \mathbf{w}_B \\ \mathbf{w}_N \end{bmatrix} = \begin{bmatrix} \mathbf{B}^{-1}\mathbf{b} \\ \mathbf{0} \end{bmatrix}$$

- the value of the objective function at the basic solution associated to basis $\mathbf{B}$ is

$$\varphi = \mathbf{c}_B^T \mathbf{B}^{-1}\mathbf{b} + \gamma$$

Geometrical interpretation of a basic solution:

a point of the space $\mathbb{R}^n$ determined by the intersection of n hyperplanes

If $\mathbf{w}_B = \mathbf{B}^{-1}\mathbf{b} \geq \mathbf{0}$: the point of the space $\mathbb{R}^n$ determined by the intersection of n hyperplanes is
a **vertex of the feasible region**, called **basic feasible solution (b.f.s.)**

Basic solution: intersection point of 2 constraint boundaries (straight lines)

$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + s_2 = 4$$

$$\begin{vmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{vmatrix} = \begin{vmatrix} 10 \\ 4 \end{vmatrix}$$

Exercise

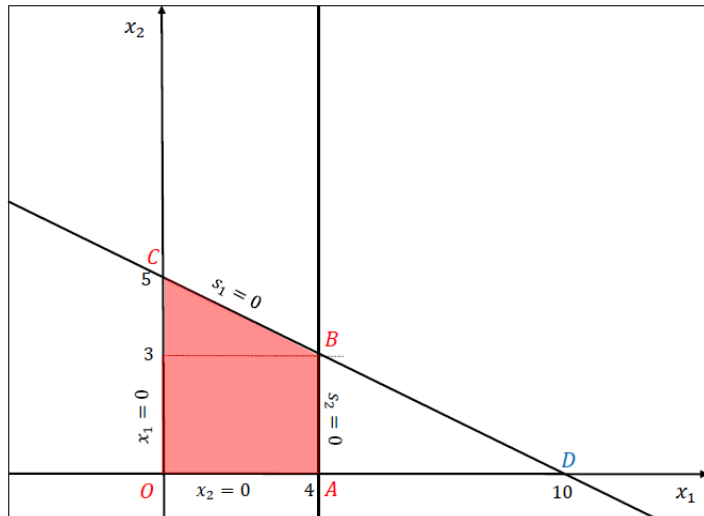
Identify all basic solutions of this problem by the graphical method

Since $\mathbf{F} \in \mathbb{R}^{2,4}$, with $\text{rank}(\mathbf{F}) = 2$,

a basis $\mathbf{B}$ consists of 2 linearly independent columns of $\mathbf{F}$;

the remaining 2 columns of $\mathbf{F}$ form the matrix $\mathbf{N}$.

We have to consider all possible ways to select 2 variables among the 4 variables (x_1, x_2, s_1, s_2) .



Example:

- let us select s_1 and s_2
- check the mutual position of the straight lines of $\mathbb{R}^2$ defined by conditions $s_1 = 0$ and $s_2 = 0$: they intersect at the point B , which is a vertex of the feasible region
- the coordinates of point B are the solution of the determined linear

system $\mathbf{B} \mathbf{w}_B = \mathbf{b}$, where $\mathbf{w}_B = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$

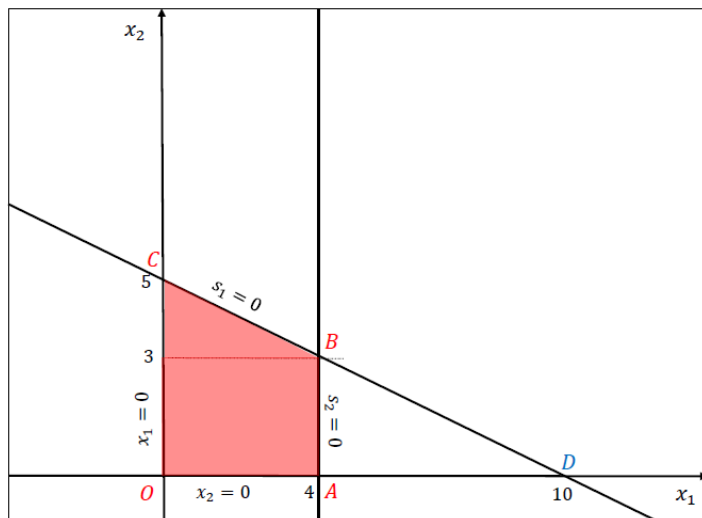
Basic solution: intersection point of 2 constraint boundaries (straight lines)

$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + s_2 = 4$$

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$$

	$w_N = \mathbf{0}$		basis $\mathbf{B}$	$w_B = \mathbf{B}^{-1}\mathbf{b}$
1	$x_1 = x_2 = 0$	vertex O	$\begin{array}{cc} s_1 & s_2 \\ \left \begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right \end{array}$	$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$



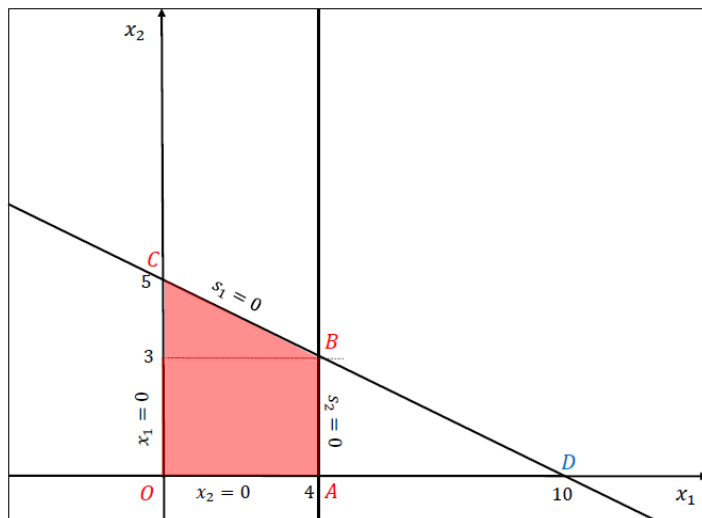
Basic solution: intersection point of 2 constraint boundaries (straight lines)

$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + s_2 = 4$$

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$$

	$w_N = \mathbf{0}$		basis $\mathbf{B}$	$w_B = \mathbf{B}^{-1}\mathbf{b}$
1	$x_1 = x_2 = 0$	vertex O	$\begin{matrix} s_1 & s_2 \\ \left \begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right \end{matrix}$	$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$
2	$x_1 = s_1 = 0$	vertex C	$\begin{matrix} x_2 & s_2 \\ \left \begin{array}{cc} 2 & 0 \\ 0 & 1 \end{array} \right \end{matrix}$	$\begin{bmatrix} x_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$



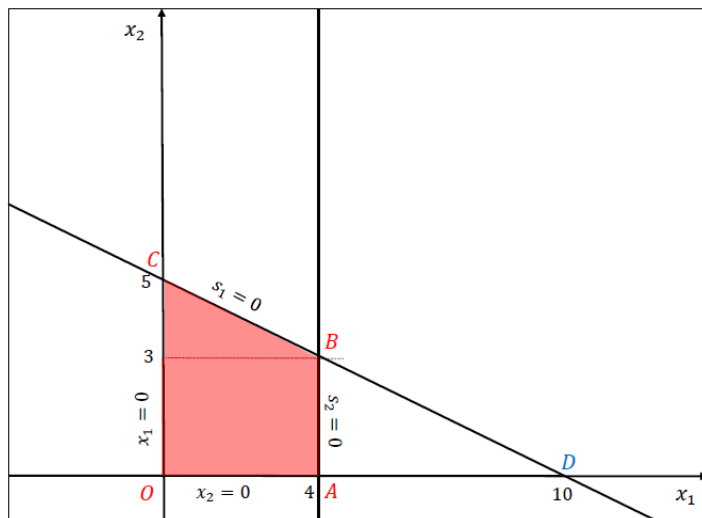
Basic solution: intersection point of 2 constraint boundaries (straight lines)

$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + s_2 = 4$$

$$\begin{vmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{vmatrix} = \begin{vmatrix} 10 \\ 4 \end{vmatrix}$$

	$w_N = \mathbf{0}$		basis $\mathbf{B}$	$w_B = \mathbf{B}^{-1}\mathbf{b}$
1	$x_1 = x_2 = 0$	vertex O	$\begin{vmatrix} s_1 & s_2 \\ 1 & 0 \\ 0 & 1 \end{vmatrix}$	$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$
2	$x_1 = s_1 = 0$	vertex C	$\begin{vmatrix} x_2 & s_2 \\ 2 & 0 \\ 0 & 1 \end{vmatrix}$	$\begin{bmatrix} x_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$
3	$x_1 = s_2 = 0$	parallel lines	$\begin{vmatrix} x_2 & s_1 \\ 2 & 1 \\ 0 & 0 \end{vmatrix}$	



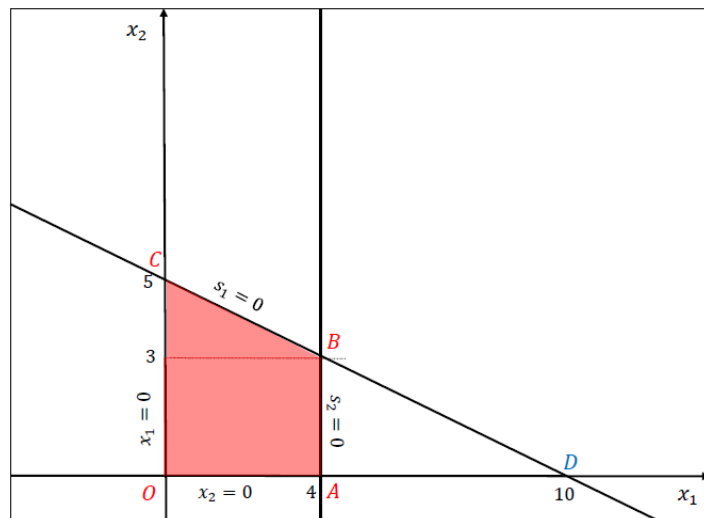
Basic solution: intersection point of 2 constraint boundaries (straight lines)

$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + s_2 = 4$$

$$\begin{vmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{vmatrix} = \begin{vmatrix} 10 \\ 4 \end{vmatrix}$$

	$w_N = \mathbf{0}$		basis $\mathbf{B}$	$w_B = \mathbf{B}^{-1}\mathbf{b}$
1	$x_1 = x_2 = 0$	vertex O	$\begin{vmatrix} s_1 & s_2 \\ 1 & 0 \\ 0 & 1 \end{vmatrix}$	$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$
2	$x_1 = s_1 = 0$	vertex C	$\begin{vmatrix} x_2 & s_2 \\ 2 & 0 \\ 0 & 1 \end{vmatrix}$	$\begin{bmatrix} x_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$
3	$x_1 = s_2 = 0$	parallel lines	$\begin{vmatrix} x_2 & s_1 \\ 2 & 1 \\ 0 & 0 \end{vmatrix}$	
4	$x_2 = s_1 = 0$	point D	$\begin{vmatrix} x_1 & s_2 \\ 1 & 0 \\ 1 & 1 \end{vmatrix}$	$\begin{bmatrix} x_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -6 \end{bmatrix}$



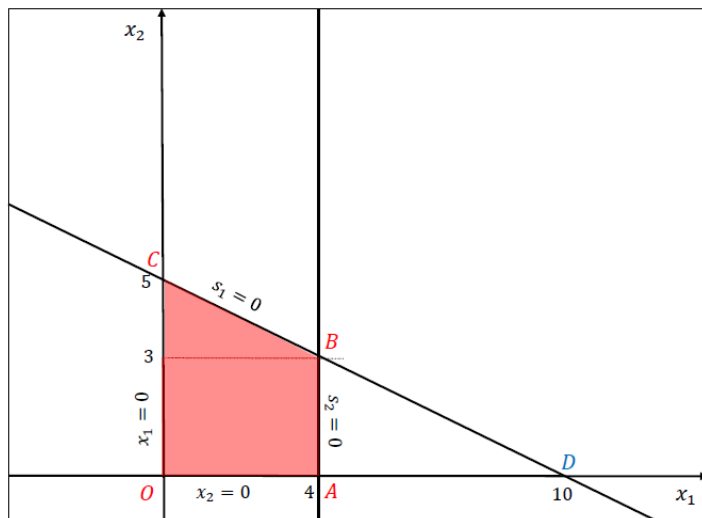
Basic solution: intersection point of 2 constraint boundaries (straight lines)

$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + s_2 = 4$$

$$\begin{vmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{vmatrix} = \begin{vmatrix} 10 \\ 4 \end{vmatrix}$$

	$w_N = 0$		basis B	$w_B = B^{-1}b$
1	$x_1 = x_2 = 0$	vertex O	$\begin{vmatrix} s_1 & s_2 \\ 1 & 0 \\ 0 & 1 \end{vmatrix}$	$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$
2	$x_1 = s_1 = 0$	vertex C	$\begin{vmatrix} x_2 & s_2 \\ 2 & 0 \\ 0 & 1 \end{vmatrix}$	$\begin{bmatrix} x_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$
3	$x_1 = s_2 = 0$	parallel lines	$\begin{vmatrix} x_2 & s_1 \\ 2 & 1 \\ 0 & 0 \end{vmatrix}$	
4	$x_2 = s_1 = 0$	point D	$\begin{vmatrix} x_1 & s_2 \\ 1 & 0 \\ 1 & 1 \end{vmatrix}$	$\begin{bmatrix} x_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -6 \end{bmatrix}$
5	$x_2 = s_2 = 0$	vertex A	$\begin{vmatrix} x_1 & s_1 \\ 1 & 1 \\ 1 & 0 \end{vmatrix}$	$\begin{bmatrix} x_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$



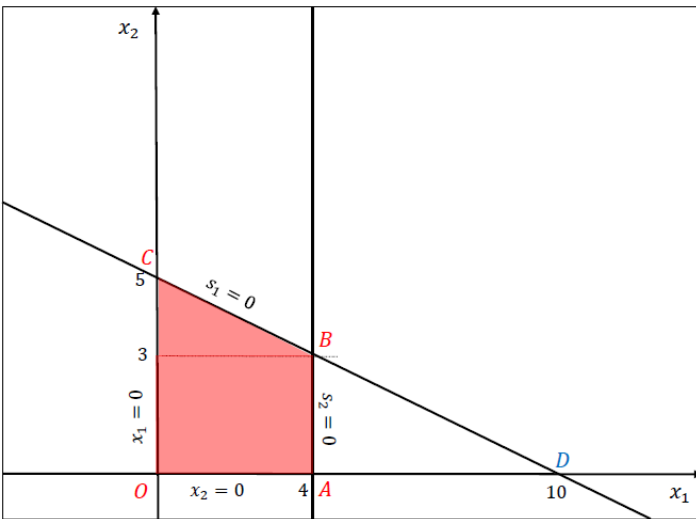
Basic solution: intersection point of 2 constraint boundaries (straight lines)

$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + s_2 = 4$$

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$$

	$w_N = \mathbf{0}$		basis $\mathbf{B}$	$w_B = \mathbf{B}^{-1}\mathbf{b}$
1	$x_1 = x_2 = 0$	vertex O	$\begin{array}{cc} s_1 & s_2 \\ \left \begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right \end{array}$	$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$
2	$x_1 = s_1 = 0$	vertex C	$\begin{array}{cc} x_2 & s_2 \\ \left \begin{array}{cc} 2 & 0 \\ 0 & 1 \end{array} \right \end{array}$	$\begin{bmatrix} x_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$
3	$x_1 = s_2 = 0$	parallel lines	$\begin{array}{cc} x_2 & s_1 \\ \left \begin{array}{cc} 2 & 1 \\ 0 & 0 \end{array} \right \end{array}$	
4	$x_2 = s_1 = 0$	point D	$\begin{array}{cc} x_1 & s_2 \\ \left \begin{array}{cc} 1 & 0 \\ 1 & 1 \end{array} \right \end{array}$	$\begin{bmatrix} x_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -6 \end{bmatrix}$
5	$x_2 = s_2 = 0$	vertex A	$\begin{array}{cc} x_1 & s_1 \\ \left \begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array} \right \end{array}$	$\begin{bmatrix} x_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$
6	$s_1 = s_2 = 0$	vertex B	$\begin{array}{cc} x_1 & x_2 \\ \left \begin{array}{cc} 1 & 2 \\ 1 & 0 \end{array} \right \end{array}$	$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$



The number of basic solutions of the underdetermined linear system $\mathbf{F} \mathbf{w} = \mathbf{b}$

(m : number of equations; $n + m$: number of unknowns) is at most

$$C_{n+m,m} = \binom{n+m}{m} = \frac{(n+m)!}{m! n!}$$

A **non-degenerate basic solution** is a solution of $\mathbf{F} \mathbf{w} = \mathbf{b}$ such that

- n components are zero
- m components are non-zero (either positive or negative)

A **degenerate basic solution** is a basic solution of $\mathbf{F} \mathbf{w} = \mathbf{b}$ for which

at least one basic variable has value zero.

- n° of zero components $> n$
- n° of non-zero components $< m$

$$\max \omega = 2x_1 + 3x_2$$

$$x_1 + 2x_2 \leq 10$$

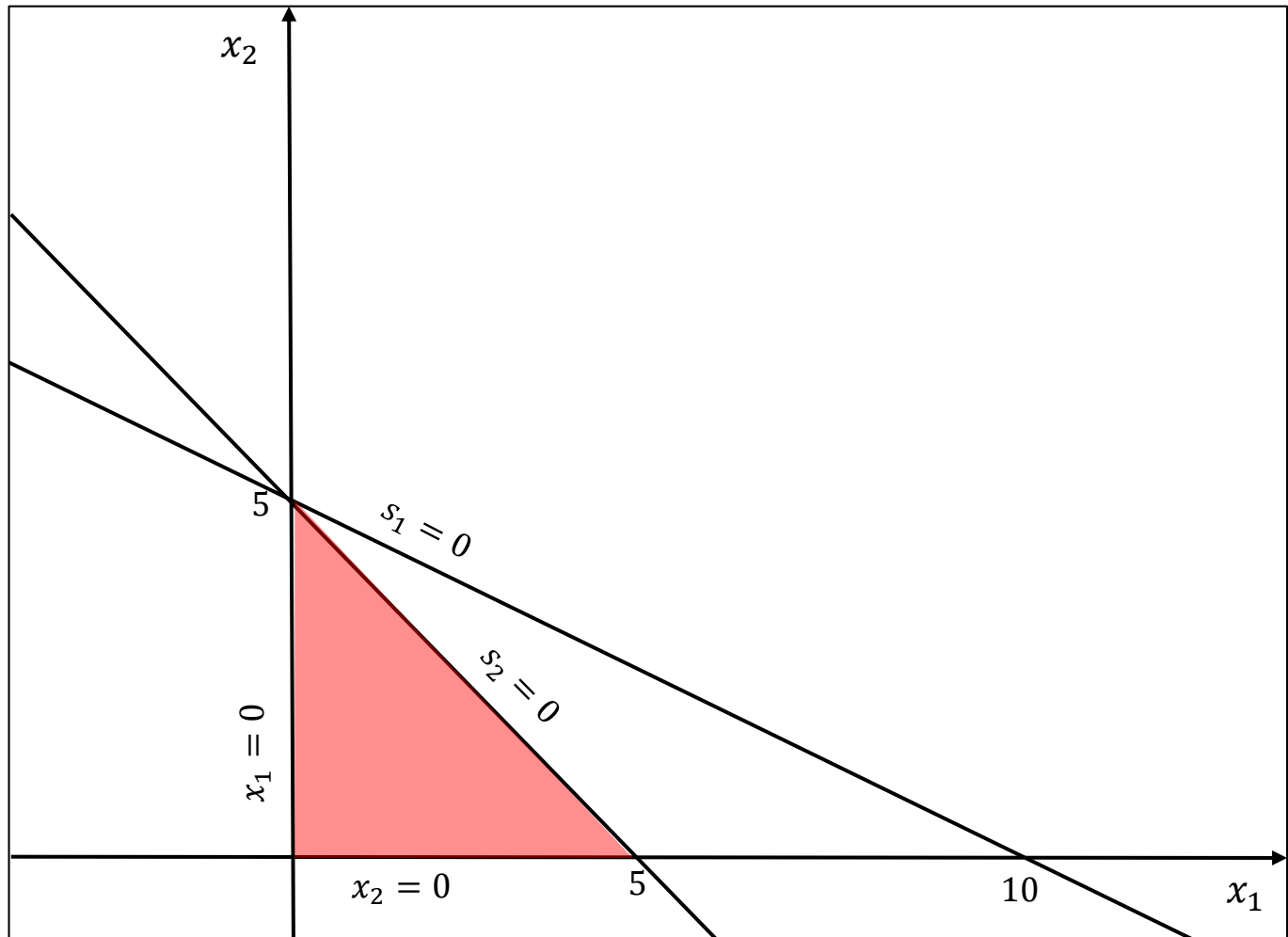
$$x_1 + x_2 \leq 5$$

$$x_1 \geq 0 \quad x_2 \geq 0$$

$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + x_2 + s_2 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

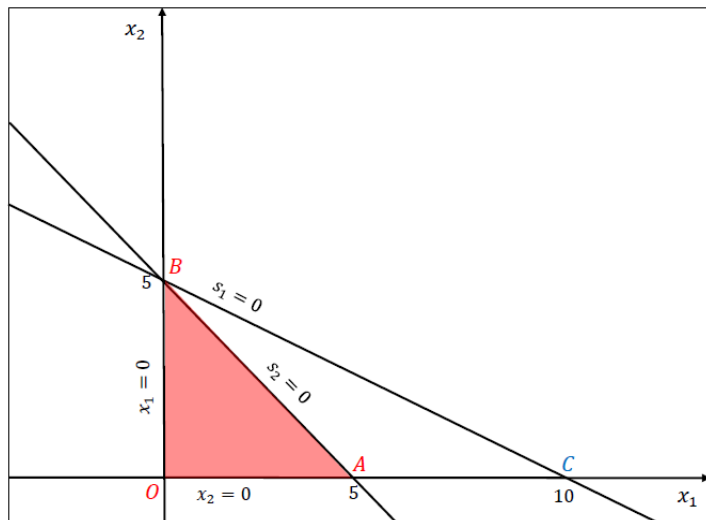


$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + x_2 + s_2 = 5$$

$$\begin{vmatrix} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{vmatrix} = \begin{vmatrix} 10 \\ 5 \end{vmatrix}$$

$$x_1, x_2, s_1, s_2 \geq 0$$



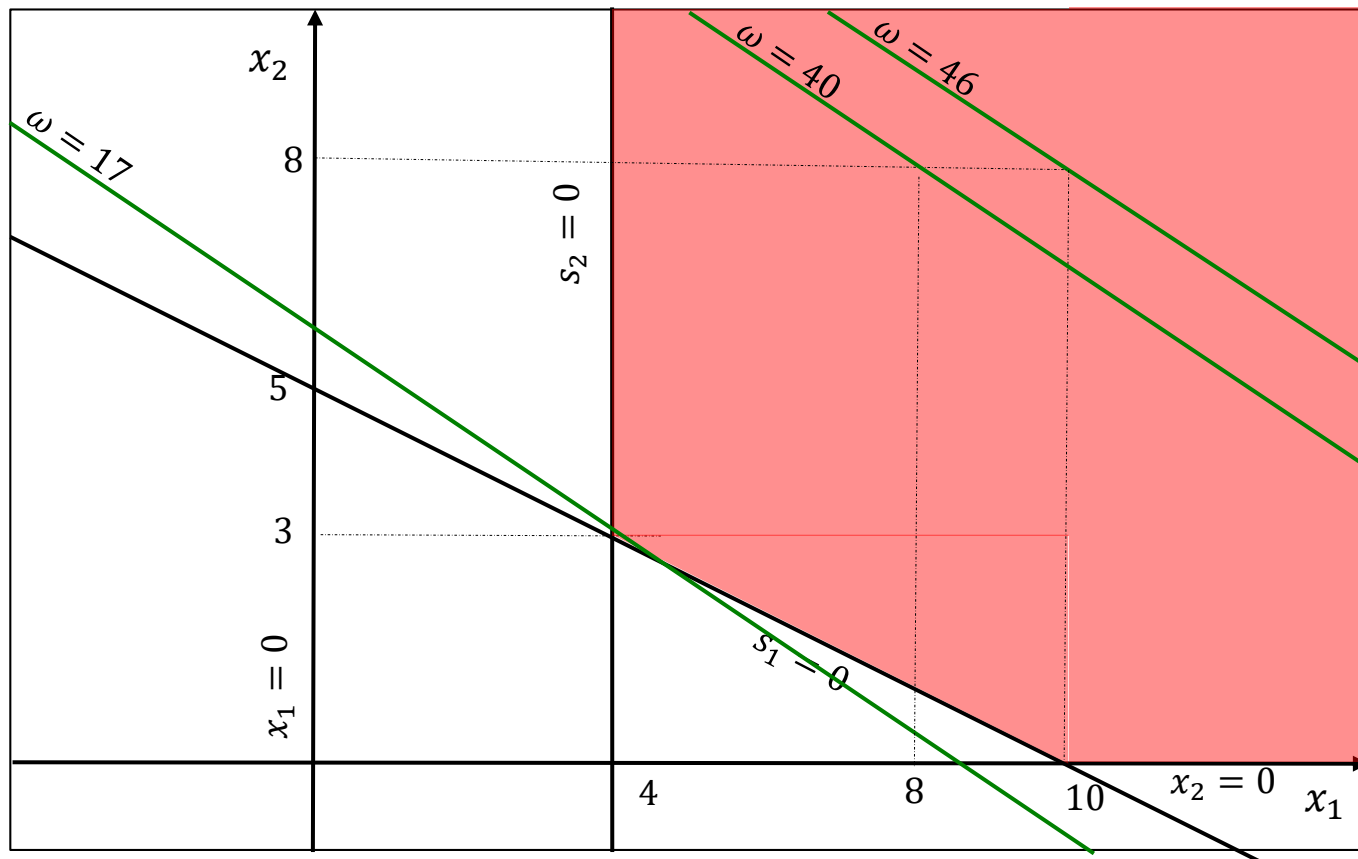
	$w_N = 0$	basis B	$w_B = B^{-1}b$	
1	$x_1 = x_2 = 0$	$\begin{vmatrix} s_1 & s_2 \\ 1 & 0 \\ 0 & 1 \end{vmatrix}$	$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$	vertex O
2	$x_1 = s_1 = 0$	$\begin{vmatrix} x_2 & s_2 \\ 2 & 0 \\ 1 & 1 \end{vmatrix}$	$\begin{bmatrix} x_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$	vertex B
3	$x_1 = s_2 = 0$	$\begin{vmatrix} x_2 & s_1 \\ 2 & 1 \\ 1 & 0 \end{vmatrix}$	$\begin{bmatrix} x_2 \\ s_1 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$	vertex B
4	$x_2 = s_1 = 0$	$\begin{vmatrix} x_1 & s_2 \\ 1 & 0 \\ 1 & 1 \end{vmatrix}$	$\begin{bmatrix} x_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} 10 \\ -5 \end{bmatrix}$	point C
5	$x_2 = s_2 = 0$	$\begin{vmatrix} x_1 & s_1 \\ 1 & 1 \\ 1 & 0 \end{vmatrix}$	$\begin{bmatrix} x_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$	vertex A
6	$s_1 = s_2 = 0$	$\begin{vmatrix} x_1 & x_2 \\ 1 & 2 \\ 1 & 1 \end{vmatrix}$	$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$	vertex B

Point $B \equiv (0, 5)$ corresponds to **3 distinct bases** of coefficient matrix F

$$B_2 = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \quad B_3 = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}, \quad B_6 = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$$

Unbounded LP problem

$$\begin{aligned}
 \max \omega &= 2x_1 + 3x_2 \\
 x_1 + 2x_2 &\geq 10 \\
 x_1 &\geq 4 \\
 x_1 \geq 0 \quad x_2 &\geq 0
 \end{aligned}$$

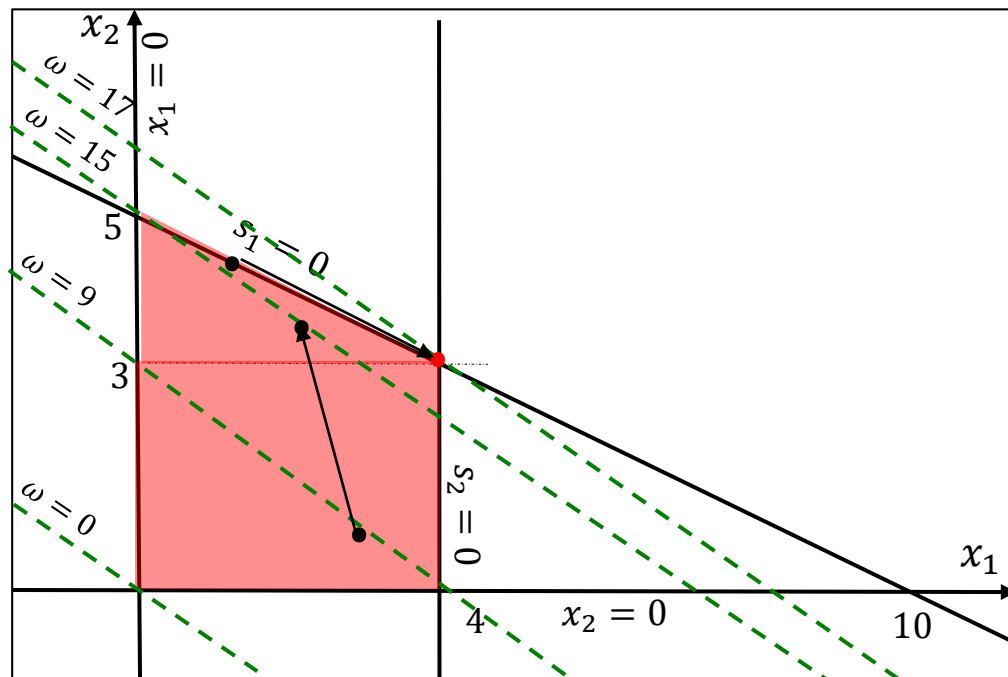


Notes

In the two-variable cases it can be observed *visually* that

- 1) if there is a unique optimal solution, such solution is a basic solution
- 2) if there are multiple (infinitely many) optimal solutions, at least one of them is a basic solution

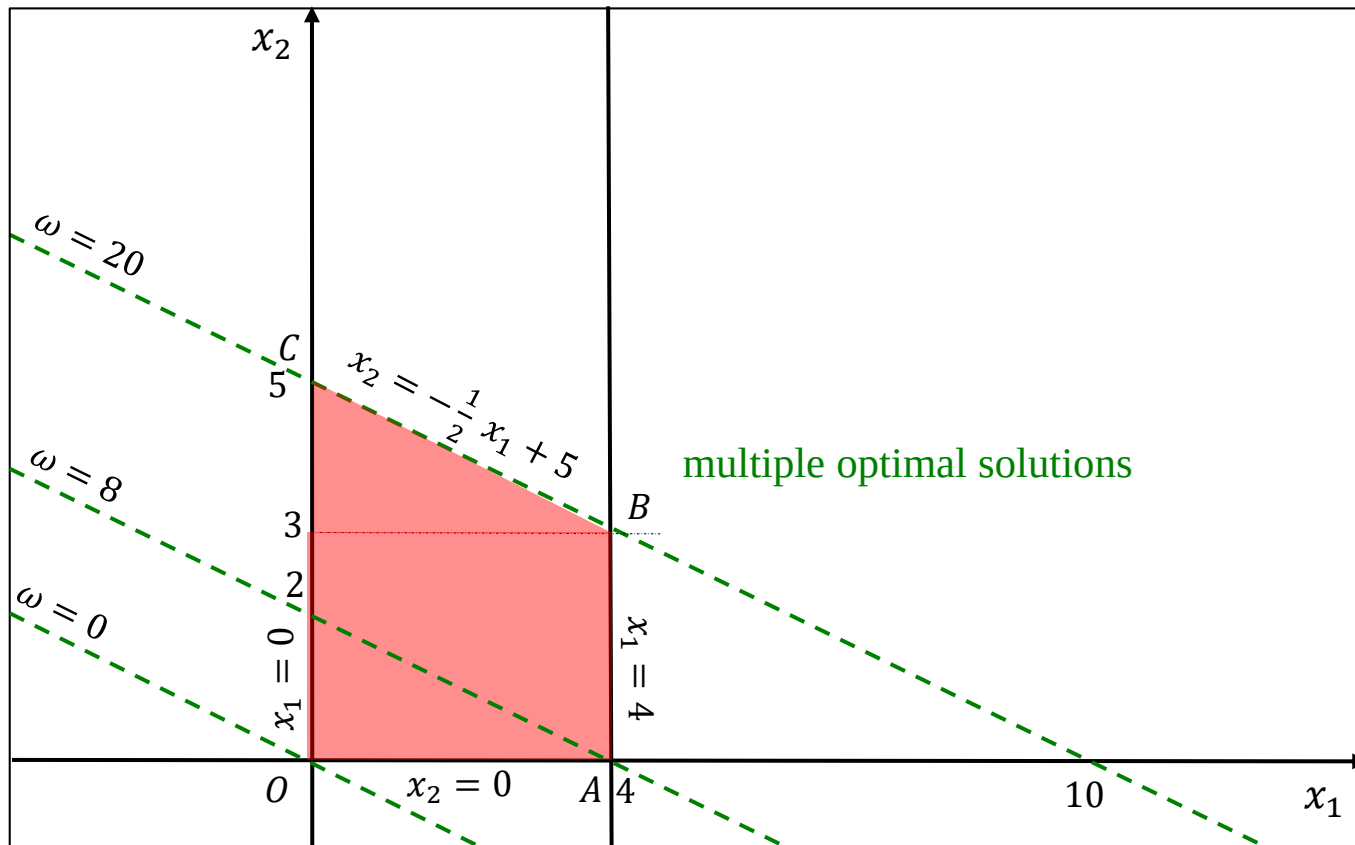
If the **optimal solution is unique**, it is a vertex because at any point that is not a vertex there is a direction leading to a point with a better objective function value



In the depicted case there are **infinitely many optimal solution**,

namely the point of the segment $\overline{BC}$ of the straight line $x_2 = -\frac{1}{2}x_1 + 5$:

the extreme points B and C are vertices.



The Fundamental Theorem of Linear Programming

Given the LP problem

$$\begin{aligned} \min \varphi = & \quad \mathbf{c}_w^T \mathbf{w} + \gamma \\ \text{s.t.} \quad & \quad \mathbf{F} \mathbf{w} = \mathbf{b} \quad \mathbf{F} \in \mathbb{R}^{m, n+m}, \text{rank}(\mathbf{F}) = m \\ & \quad \mathbf{w} \geq 0 \end{aligned}$$

- 1) if feasible solutions exist, then at least one of them is basic.
- 2) if optimal feasible solutions exist, then at least one of them is basic.

Before proving the theorem, let us highlight the consequences of these statements:

The statement in Part 2

“ if optimal feasible solutions exist, then at least one of them is basic ”

is equivalent to the statement

“ if the feasible region is non-empty and the problem is not unbounded,
there exists an optimal basic feasible solution ”

Consequence of Part 2

the optimal solution can be sought in the subset of the basic feasible solutions

Such subset has **finite cardinality**

(while in general the feasible region contains infinitely many solutions)

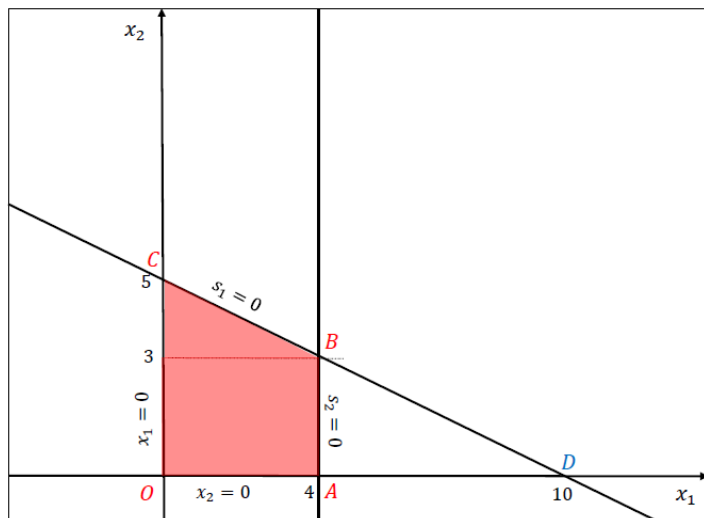
A simple (but inefficient) approach: the **enumeration** algorithm

$$\max \omega = 2x_1 + 3x_2$$

$$x_1 + 2x_2 + s_1 = 10$$

$$x_1 + s_2 = 4$$

$$\begin{vmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{vmatrix} = \begin{vmatrix} 10 \\ 4 \end{vmatrix}$$



	$w_N = \mathbf{0}$		basis $\mathbf{B}$	$w_B = \mathbf{B}^{-1}\mathbf{b}$	ω
1	$x_1 = x_2 = 0$	vertex O	$\begin{vmatrix} s_1 & s_2 \\ 1 & 0 \\ 0 & 1 \end{vmatrix}$	$\begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$	0
2	$x_1 = s_1 = 0$	vertex C	$\begin{vmatrix} x_2 & s_2 \\ 2 & 0 \\ 0 & 1 \end{vmatrix}$	$\begin{bmatrix} x_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$	15
3	$x_1 = s_2 = 0$	parallel lines	$\begin{vmatrix} x_2 & s_1 \\ 2 & 1 \\ 0 & 0 \end{vmatrix}$		
4	$x_2 = s_1 = 0$	point D	$\begin{vmatrix} x_1 & s_2 \\ 1 & 0 \\ 1 & 1 \end{vmatrix}$	$\begin{bmatrix} x_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -6 \end{bmatrix}$	(20)
5	$x_2 = s_2 = 0$	vertex A	$\begin{vmatrix} x_1 & s_1 \\ 1 & 1 \\ 1 & 0 \end{vmatrix}$	$\begin{bmatrix} x_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$	8
6	$s_1 = s_2 = 0$	vertex B	$\begin{vmatrix} x_1 & x_2 \\ 1 & 2 \\ 1 & 0 \end{vmatrix}$	$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$	17

A more efficient approach: the **simplex algorithm (optimization phase)**

The **simplex algorithm** is an **iterative** algorithm.

We describe the structure of the generic iteration i for solving a minimization problem.

At the **beginning of iteration i** the following is given:

- the canonical form of $\mathbf{F} \mathbf{w} = \mathbf{b}$ associated to the current basis $\mathbf{B}$,
i.e. the expression of each basic variable in terms of the non-basic variables

$$\mathbf{w}_B + \mathbf{B}^{-1} \mathbf{N} \mathbf{w}_N = \mathbf{B}^{-1} \mathbf{b} \geq \mathbf{0}$$

- the expression of the objective variable in terms of the non-basic variables

$$\varphi = \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b} + \gamma - \mathbf{r}_N^T \mathbf{w}_N$$

$\mathbf{r}_N^T$: vector of relative costs of non-basic variables

The current basic feasible solution is

$$\mathbf{w}_i = \begin{bmatrix} \mathbf{w}_B \\ \mathbf{w}_N \end{bmatrix} = \begin{bmatrix} \mathbf{B}^{-1} \mathbf{b} \\ \mathbf{0} \end{bmatrix}$$

with objective function value

$$\varphi_i = \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b} + \gamma$$

Step 1: optimality test

By using the expression of the objective variable in terms of the non-basic variables

$$\varphi = \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b} + \gamma - \mathbf{r}_N^T \mathbf{w}_N$$

we can determine the objective value $\tilde{\varphi}$ corresponding to a vector $\tilde{\mathbf{w}}_N \geq \mathbf{0}$,

where at least one non-basic variable takes a positive value.

If $\mathbf{r}_N^T \leq \mathbf{0}^T$ (i.e. the relative costs of all non-basic variables are nonpositive),

then the dot product $\mathbf{r}_N^T \tilde{\mathbf{w}}_N \leq 0$ for any vector $\tilde{\mathbf{w}}_N \geq \mathbf{0}$ and

$$\tilde{\varphi} = \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b} + \gamma - \mathbf{r}_N^T \tilde{\mathbf{w}}_N \geq \varphi_i = \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b} + \gamma$$

The current b.f.s. is optimal and the algorithm is stopped.

If the relative cost of the non-basic variable $\mathbf{w}_k$ is positive ($r_k > 0$) and the remaining

non-basic variables are zero, then $\tilde{\varphi} = \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b} + \gamma - r_k \mathbf{w}_k < \varphi_i = \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b} + \gamma$,

i.e. the current b.f.s. is non optimal.

Step 2: **determine a new b.f.s.**

- **adjacent to the current one**
- **with objective function value not greater than the one at the current b.f.s.**

If there is more than one non-basic variable with a positive relative cost,
select the one with the largest relative cost and let w_k denote the selected non-basic variable.

By using the canonical form (where each basic variable expressed in terms of the non-basic variables)

$$w_B + B^{-1}N w_N = B^{-1}b$$

we can **impose the feasibility conditions**

$$\tilde{w}_B = B^{-1}b - B^{-1}f_k \tilde{w}_k \geq 0$$

by which we determine the adjacent b.f.s. (and therefore the new basis $\tilde{B}$)

Step 3: **determine the canonical form of the new b.f.s.**

as well as the expression of φ in terms of the new vector $\tilde{\mathbf{w}}_N$ of non-basic variables.

To obtain the canonical form of the new b.f.s.

- we **do not** premultiply both sides of $\mathbf{F} \mathbf{w} = \mathbf{b}$ by the inverse $\tilde{\mathbf{B}}^{-1}$ of the new basis $\tilde{\mathbf{B}}$
- we perform, instead, $m + 1$ «row elementary operations»

In other words, the Simplex method generates a sequence of b.f.s.

with objective function value monotonically non increasing,

until an optimal b.f.s. is determined.

The statement in Part 1

“ if feasible solutions exist, then at least one of them is basic ”

is equivalent to the statement

“ if all basic solutions of $\mathbf{F} \mathbf{w} = \mathbf{b}$ are infeasible ($\mathbf{w}_B \not\geq \mathbf{0}$), no feasible solutions exist ”

Consequence of Part 1

Consider the LP problems for which the zero values of the decision variables ($\mathbf{x} = \mathbf{0}$) violate at least one functional constraint.

To determine whether the feasible region is empty (i.e. whether the LP problem is feasible or infeasible), we can examine all basic solutions:

if at least one basic solution is feasible, the LP problem is feasible,

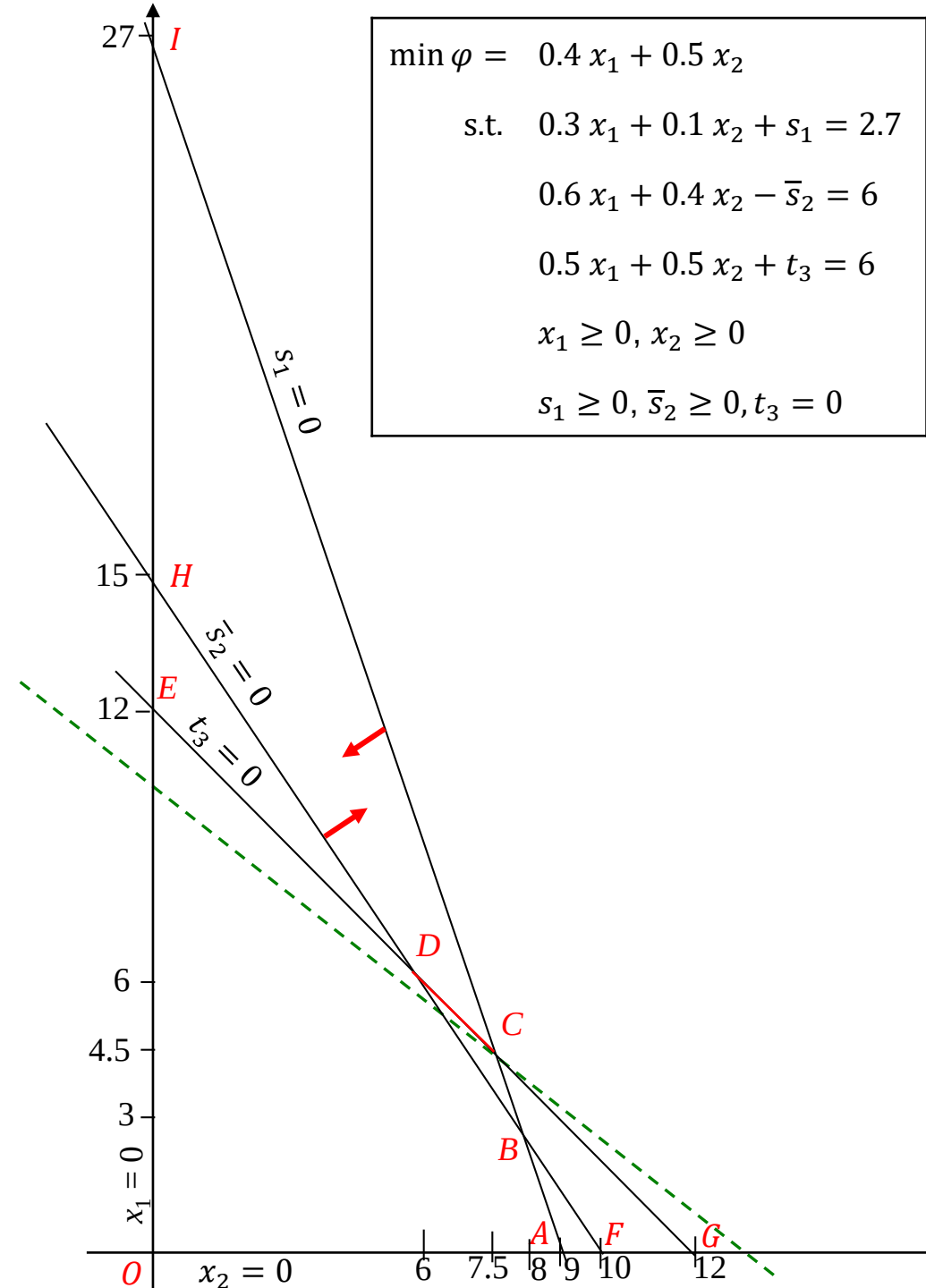
if none are feasible, the LP problem is infeasible.

The case when $\mathbf{x} = \mathbf{0}$ do not belong to the feasible region

The **enumeration** algorithm: a simple (but inefficient) approach
(two examples: a feasible problem and an infeasible problem)

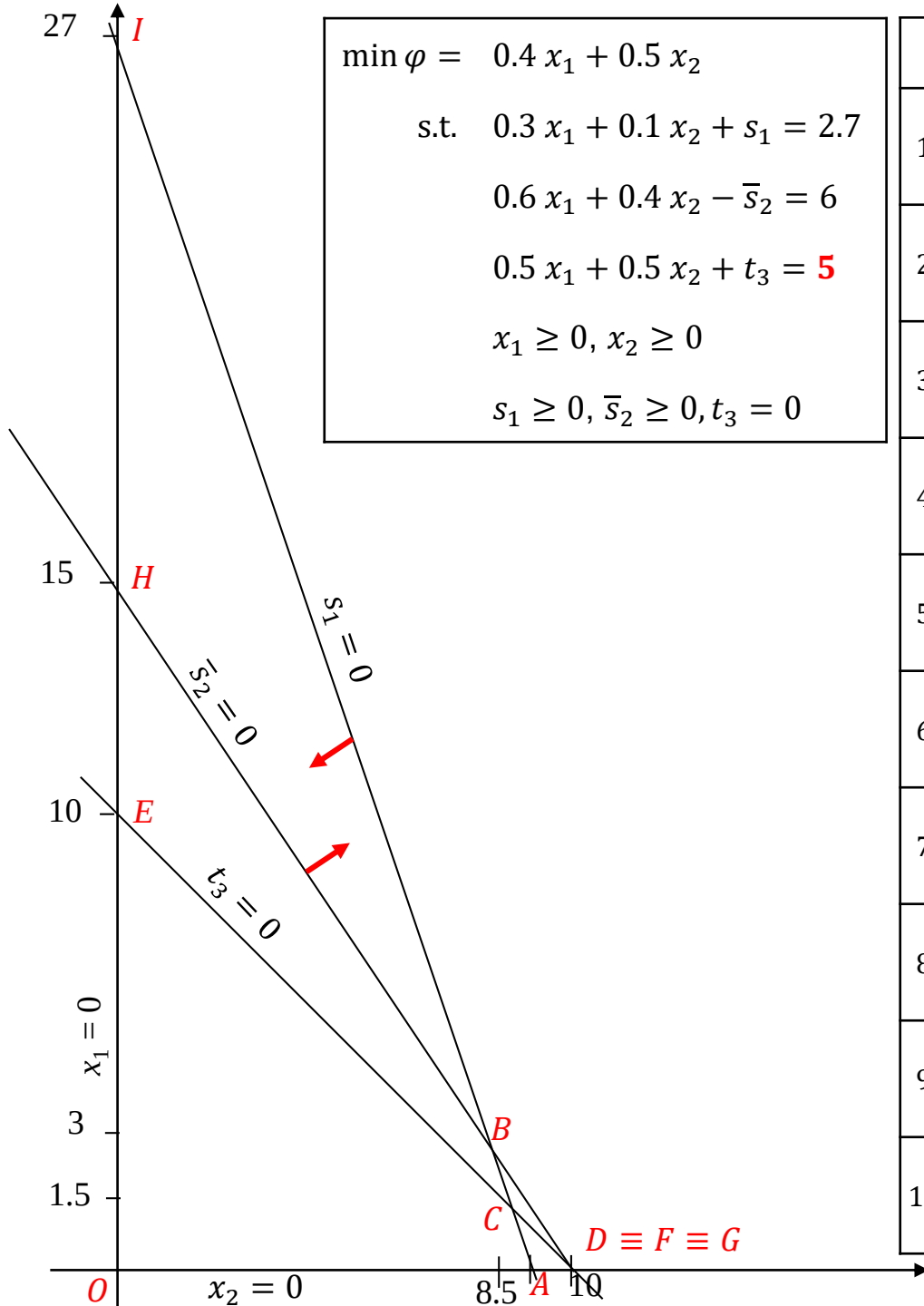
versus

The **simplex algorithm (feasibility phase)**: a more efficient approach

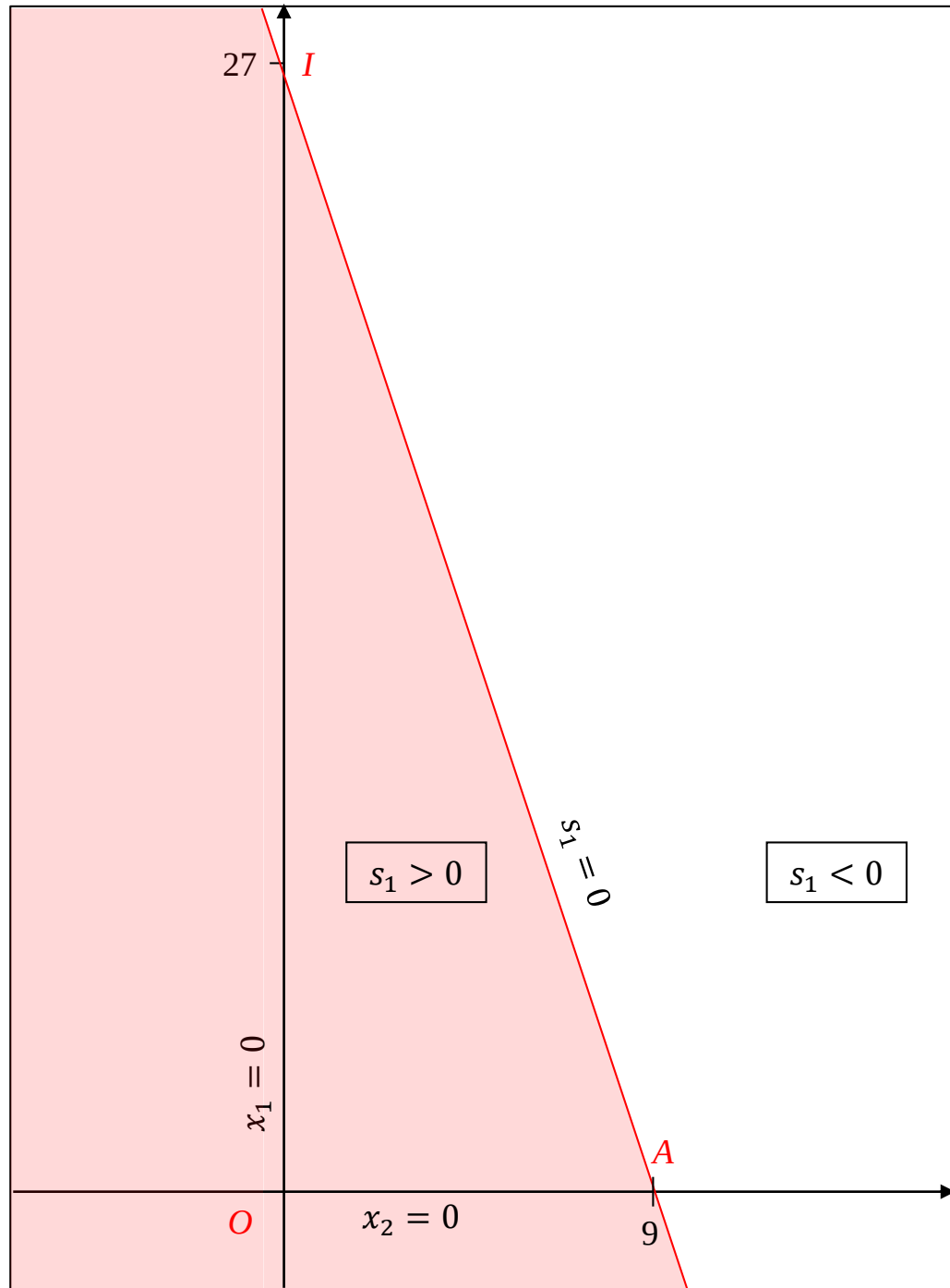


$$\begin{aligned} \min \varphi = & 0.4 x_1 + 0.5 x_2 \\ \text{s.t. } & 0.3 x_1 + 0.1 x_2 + s_1 = 2.7 \\ & 0.6 x_1 + 0.4 x_2 - \bar{s}_2 = 6 \\ & 0.5 x_1 + 0.5 x_2 + t_3 = 6 \\ & x_1 \geq 0, x_2 \geq 0 \\ & s_1 \geq 0, \bar{s}_2 \geq 0, t_3 = 0 \end{aligned}$$

	$\mathbf{w}_N = \mathbf{0}$	point in $\mathbb{R}^2$	$\mathbf{w}_B = \mathbf{B}^{-1}\mathbf{b}$
1	$x_1 = x_2 = 0$	O	$\begin{bmatrix} s_1 \\ \bar{s}_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} 2.7 \\ -6 \\ 6 \end{bmatrix}$
2	$x_1 = s_1 = 0$	I	$\begin{bmatrix} x_2 \\ \bar{s}_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} 27 \\ 4.8 \\ -7.5 \end{bmatrix}$
3	$x_1 = \bar{s}_2 = 0$	H	$\begin{bmatrix} x_2 \\ s_1 \\ t_3 \end{bmatrix} = \begin{bmatrix} 15 \\ 1.2 \\ -1.5 \end{bmatrix}$
4	$x_1 = t_3 = 0$	E	$\begin{bmatrix} x_2 \\ s_1 \\ \bar{s}_2 \end{bmatrix} = \begin{bmatrix} 12 \\ 1.5 \\ -1.2 \end{bmatrix}$
5	$x_2 = s_1 = 0$	A	$\begin{bmatrix} x_1 \\ \bar{s}_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} 9 \\ -0.6 \\ 1.5 \end{bmatrix}$
6	$x_2 = \bar{s}_2 = 0$	F	$\begin{bmatrix} x_1 \\ s_1 \\ t_3 \end{bmatrix} = \begin{bmatrix} 10 \\ -0.3 \\ 1 \end{bmatrix}$
7	$x_2 = t_3 = 0$	G	$\begin{bmatrix} x_1 \\ s_1 \\ \bar{s}_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -0.9 \\ 1.2 \end{bmatrix}$
8	$s_1 = \bar{s}_2 = 0$	B	$\begin{bmatrix} x_1 \\ x_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \\ 0.5 \end{bmatrix}$
9	$s_1 = t_3 = 0$	C	$\begin{bmatrix} x_1 \\ x_2 \\ \bar{s}_2 \end{bmatrix} = \begin{bmatrix} 7.5 \\ 4.5 \\ 0.3 \end{bmatrix}$
10	$\bar{s}_2 = t_3 = 0$	D	$\begin{bmatrix} x_1 \\ x_2 \\ s_1 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ 0.3 \end{bmatrix}$

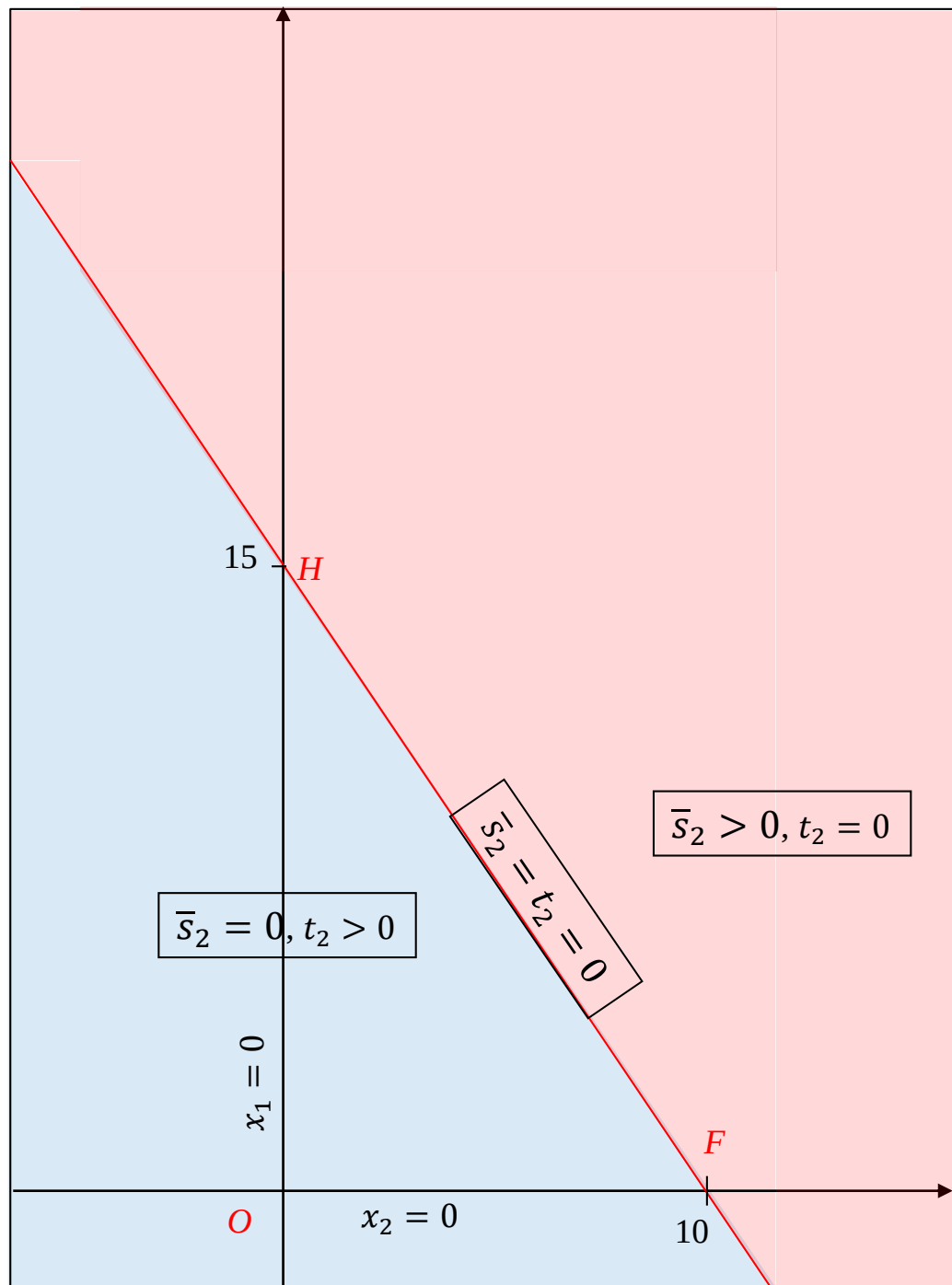


	$\mathbf{w}_N = \mathbf{0}$	point in $\mathbb{R}^2$	$\mathbf{w}_B = \mathbf{B}^{-1}\mathbf{b}$
1	$x_1 = x_2 = 0$	O	$\begin{bmatrix} \bar{s}_1 \\ \bar{s}_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} 2.7 \\ -6 \\ 5 \end{bmatrix}$
2	$x_1 = s_1 = 0$	I	$\begin{bmatrix} x_2 \\ \bar{s}_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} 27 \\ 4.8 \\ -8.5 \end{bmatrix}$
3	$x_1 = \bar{s}_2 = 0$	H	$\begin{bmatrix} x_2 \\ s_1 \\ t_3 \end{bmatrix} = \begin{bmatrix} 15 \\ 1.2 \\ -2.5 \end{bmatrix}$
4	$x_1 = t_3 = 0$	E	$\begin{bmatrix} x_2 \\ s_1 \\ \bar{s}_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 1.7 \\ -2 \end{bmatrix}$
5	$x_2 = s_1 = 0$	A	$\begin{bmatrix} x_1 \\ \bar{s}_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} 9 \\ -0.6 \\ 0.5 \end{bmatrix}$
6	$x_2 = \bar{s}_2 = 0$	F	$\begin{bmatrix} x_1 \\ s_1 \\ t_3 \end{bmatrix} = \begin{bmatrix} 10 \\ -0.3 \\ 0 \end{bmatrix}$
7	$x_2 = t_3 = 0$	G	$\begin{bmatrix} x_1 \\ s_1 \\ \bar{s}_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -0.3 \\ 0 \end{bmatrix}$
8	$s_1 = \bar{s}_2 = 0$	B	$\begin{bmatrix} x_1 \\ x_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \\ -0.5 \end{bmatrix}$
9	$s_1 = t_3 = 0$	C	$\begin{bmatrix} x_1 \\ x_2 \\ \bar{s}_2 \end{bmatrix} = \begin{bmatrix} 8.5 \\ 1.5 \\ -0.3 \end{bmatrix}$
10	$\bar{s}_2 = t_3 = 0$	D	$\begin{bmatrix} x_1 \\ x_2 \\ s_1 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ -0.3 \end{bmatrix}$



$$0.3 x_1 + 0.1 x_2 + s_1 = 2.7$$

$$s_1 \geq 0$$



$$0.6 x_1 + 0.4 x_2 - \bar{s}_2 + t_2 = 6$$

$$\bar{s}_2 \geq 0$$

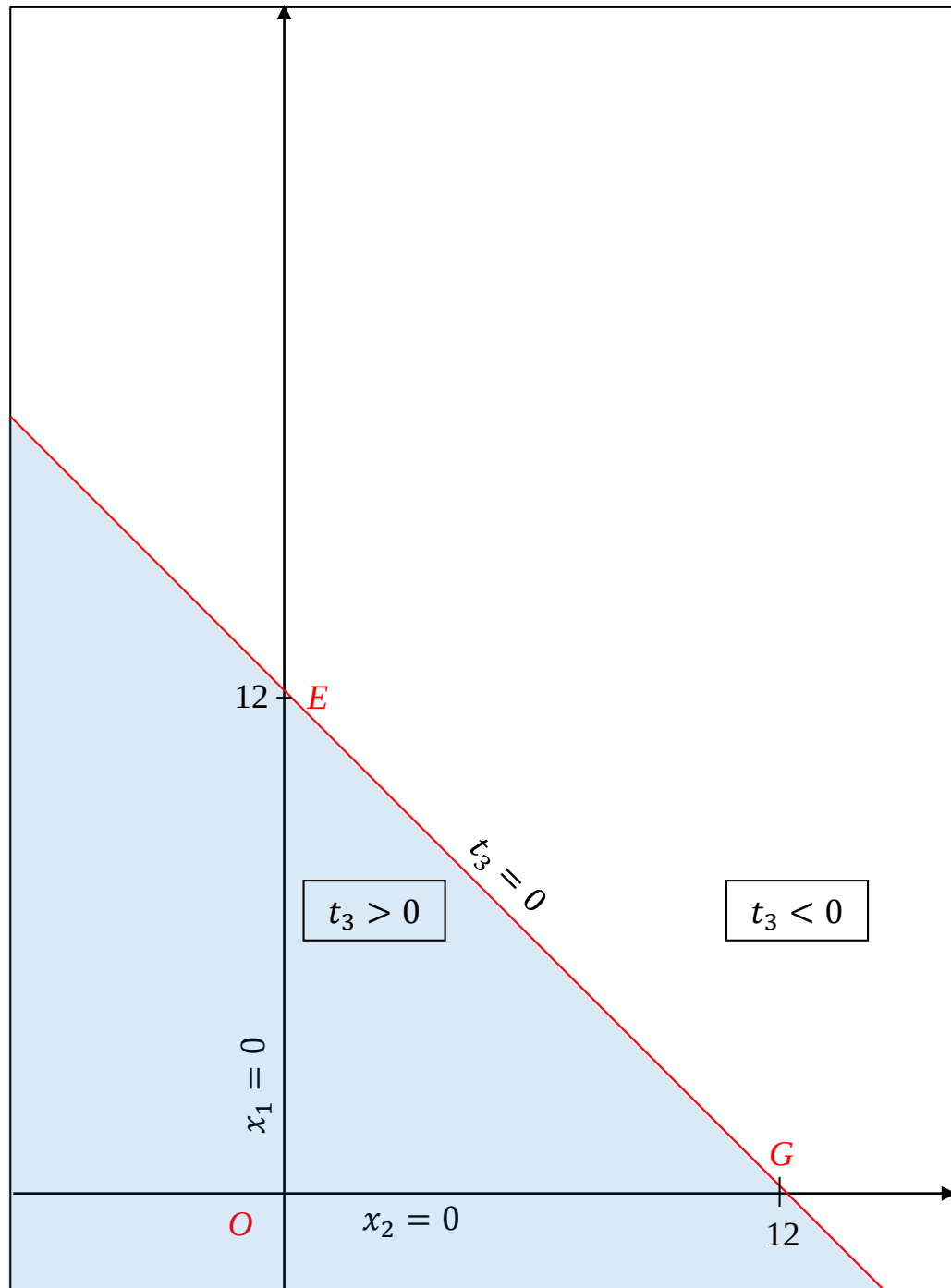
$$t_2 \geq 0$$

which corresponds to

$$0.6 x_1 + 0.4 x_2 - u_2 = 6$$

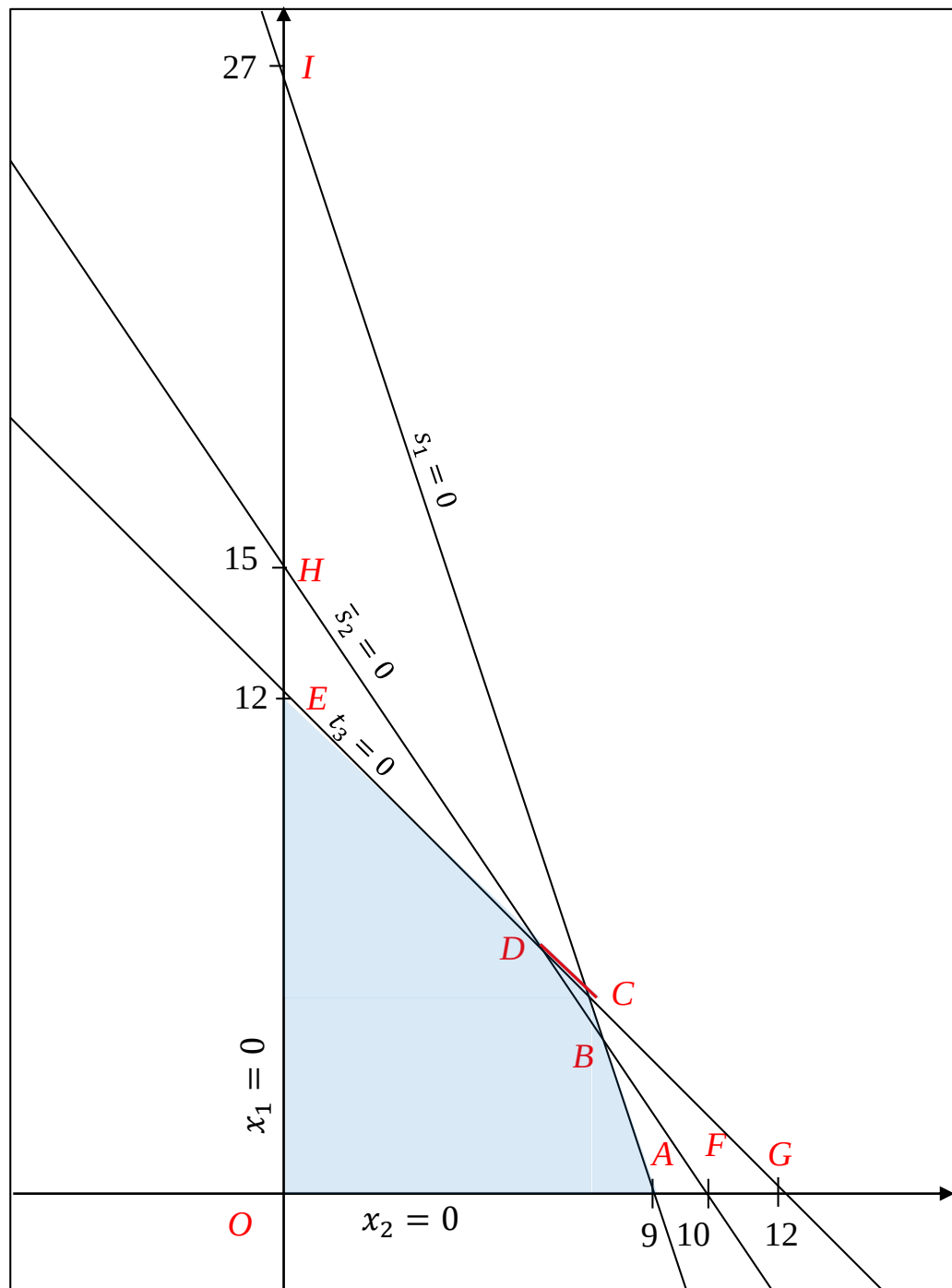
$$u_2 = \bar{s}_2 - t_2 \text{ free variable}$$

- $u_2 > 0$, for $\bar{s}_2 > 0$ and $t_2 = 0$:
point (x_1, x_2) belongs to the
half-plane **above** the line $\bar{s}_2 = 0$
- $u_2 < 0$, for $\bar{s}_2 = 0$ and $t_2 > 0$:
point (x_1, x_2) belongs to the
half-plane **below** the line $\bar{s}_2 = 0$



$$0.5 x_1 + 0.5 x_2 + t_3 = 6$$

$$t_3 \geq 0$$



Feasible region of the
auxiliary problem

$$0.3 x_1 + 0.1 x_2 + s_1 = 2.7$$

$$0.6 x_1 + 0.4 x_2 - \bar{s}_2 + t_2 = 6$$

$$0.5 x_1 + 0.5 x_2 + t_3 = 6$$

$$x_1 \geq 0, x_2 \geq 0, s_1 \geq 0, \bar{s}_2 \geq 0, t_2 \geq 0, t_3 \geq 0$$

Il problema ausiliario ha una regione ammissibile modificata

- accettando valori $t_2 > 0$ si aggiunge il semipiano sotto la retta $\bar{s}_2 = 0$
- accettando valori $t_3 > 0$ si aggiunge il semipiano sotto la retta $t_3 = 0$

La regione ammissibile del problema originario è invece

$$0.3 x_1 + 0.1 x_2 + s_1 = 2.7$$

$$0.6 x_1 + 0.4 x_2 - \bar{s}_2 + t_2 = 6$$

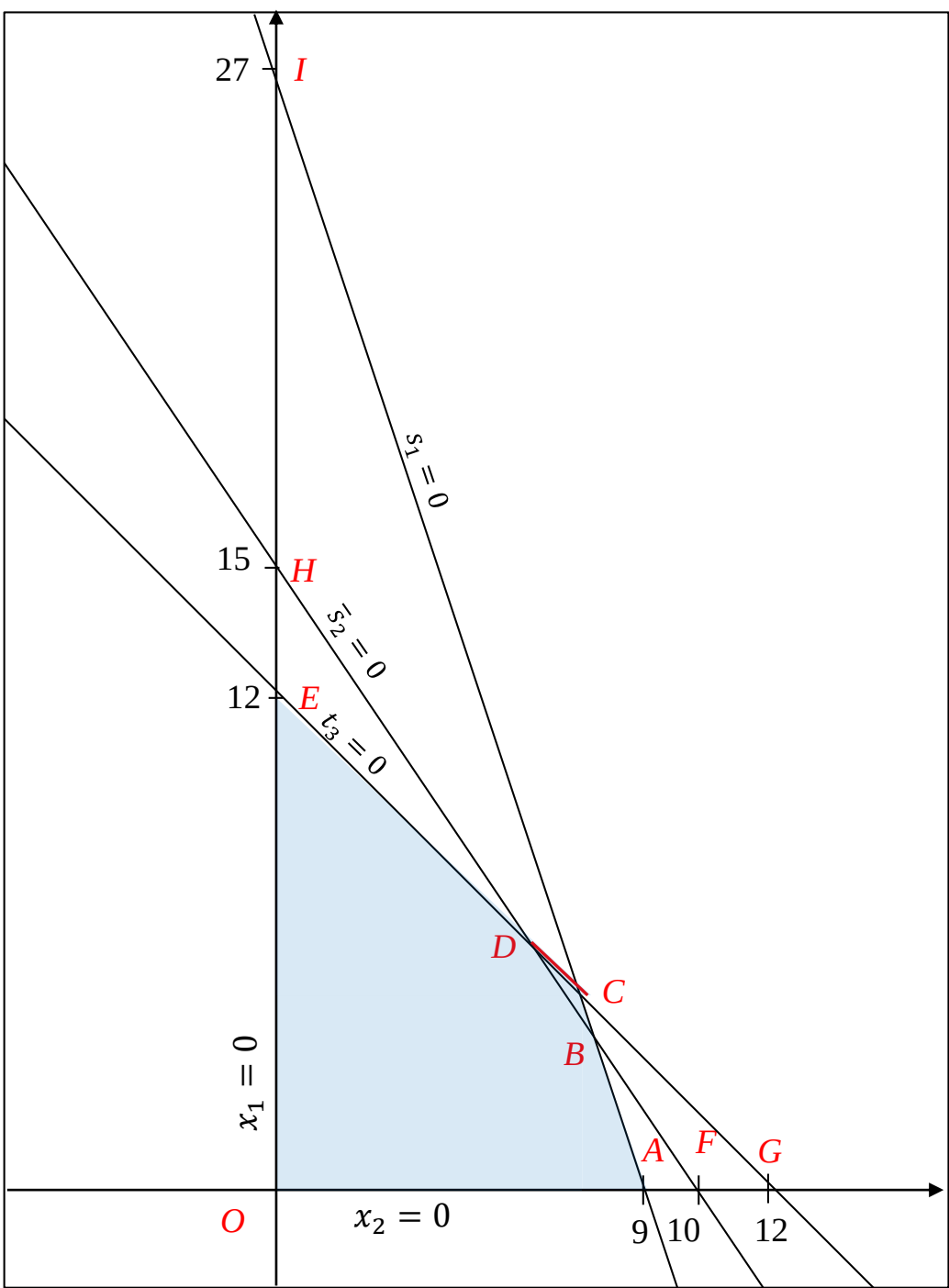
$$0.5 x_1 + 0.5 x_2 + t_3 = 6$$

$$x_1 \geq 0, x_2 \geq 0, s_1 \geq 0, \bar{s}_2 \geq 0, t_2 = 0, t_3 = 0$$

Il problema originario e il problema ausiliario hanno le stesse soluzioni di base:

una soluzione del sistema lineare tale che $x_1 \geq 0, x_2 \geq 0, s_1 \geq 0, \bar{s}_2 \geq 0$

- è **ammissibile** per il problema **ausiliario** se $t_2 \geq 0, t_3 \geq 0$
- è **ammissibile** per il problema **originario** se $t_2 = 0, t_3 = 0$



Basic feasible solutions of the auxiliary problem

	x_1	x_2	t_2	t_3	$\Psi = t_2 + t_3$
O	0	0	6	6	12
A	9	0	$\frac{3}{5}$	$\frac{3}{2}$	$\frac{21}{10}$
B	8	3	0	$\frac{1}{2}$	$\frac{1}{2}$
C	$\frac{15}{2}$	$\frac{9}{2}$	$\max\left\{0, -\frac{3}{10}\right\} = 0$	0	0
D	6	6	0	0	0
E	0	12	$\frac{6}{5}$	0	$\frac{6}{5}$

with

$$t_2 = \max\left\{0, \left(6 - \frac{3}{5} x_1 - \frac{2}{5} x_2\right)\right\}$$

$$t_3 = \max\left\{0, \left(6 - \frac{1}{2} x_1 - \frac{1}{2} x_2\right)\right\}$$

I punti B e D

I punti B e D non sono **vertici** della regione ammissibile del problema ausiliario, ma sono intersezione di frontiere di vincoli e quindi vengono comunque considerati

La loro rappresentazione algebrica è disponibile, perché nel sistema dei vincoli funzionali è presente il vincolo

$$0.6 x_1 + 0.4 x_2 - u_2 = 6 \quad u_2 \geq 0$$

Punto B : intersezione delle frontiere dei vincoli funzionali primo e **secondo**

Punto D : intersezione delle frontiere dei vincoli funzionali **secondo** e terzo

Definiamo la funzione obiettivo del problema ausiliario $\Psi = t_2 + t_3$:

tale funzione obiettivo assegna un valore Ψ di **inammissibilità** ad ogni s.b.a. del problema ausiliario.

Determiniamo la s.b.a. del problema ausiliario in cui il valore di Ψ è **minimo**:

$$\min \Psi = t_2 + t_3$$

Il problema è sempre inferiormente limitato, perché $t_2 \geq 0, t_3 \geq 0$

- se $\Psi = 0$: la s.b.a.o. del problema ausiliario è s.b.a. per il problema originario
- se $\Psi > 0$: la s.b.a.o. del problema ausiliario è una s.b. non è ammissibile per il problema originario

Della variabile obiettivo ausiliaria Ψ si deve determinare il minimo:

- se il minimo è $\Psi > 0$,

nessuna s.b.a. del problema ausiliario è ammissibile per il problema originario;

poiché abbiamo dimostrato che non esistono s.b.a. del problema originario,

per il TFPL non esistono s.a. per il problema originario

- se il minimo è $\Psi = 0$,

la s.b.a. ottima del problema ausiliario è ammissibile per il problema originario:

tale s.b.a. è utilizzata come s.b.a. iniziale nella fase di ricerca della s.b.a. ottima

del problema originario.

FASE 1 : ricerca di una s.b.a.

$$\min \Psi = t_2 + t_3$$

$$\text{s.a} \quad 0.3 x_1 + 0.1 x_2 + s_1 = 2.7$$

$$0.6 x_1 + 0.4 x_2 - \bar{s}_2 + t_2 = 6$$

$$0.5 x_1 + 0.5 x_2 + t_3 = 6$$

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ	
$\frac{3}{10}$	$\frac{1}{10}$	1	0	0	0	0	$\frac{27}{10}$
$\frac{3}{5}$	$\frac{2}{5}$	0	-1	1	0	0	6
$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	1	0	6
0	0	0	0	-1	-1	1	0

Base formata dalle colonne delle variabili s_1 , t_2 e t_3 :

per ottenere la forma canonica associata a questa base,

anche la variabile obiettivo Ψ deve essere espressa in termini delle sole v.n.b. x_1 , x_2 e $\bar{s}_2$ 71

Espressione della v.o. in termini delle sole v.n.b.

$$\Psi = t_2 + t_3 = \underbrace{\left(6 - \frac{3}{5}x_1 - \frac{2}{5}x_2 + \bar{s}_2\right)}_{t_2} + \underbrace{\left(6 - \frac{1}{2}x_1 - \frac{1}{2}x_2\right)}_{t_3}$$

da cui $\frac{11}{10}x_1 + \frac{9}{10}x_2 - \bar{s}_2 + \Psi = 12$

Questa operazione può anche essere espressa come la combinazione lineare di righe

$$\bar{R}_0 = R_0 + R_2 + R_3$$

	x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ	
R_0	0	0	0	0	-1	-1	1	0
$+ R_2$	$\frac{3}{5}$	$\frac{2}{5}$	0	-1	1	0	0	6
$+ R_3$	$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	1	0	6
$= \bar{R}_0$	$\frac{11}{10}$	$\frac{9}{10}$	0	-1	0	0	1	12

Forma canonica associata alla base formata dalle colonne delle variabili s_1, t_2, t_3

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ	
$\frac{3}{10}$	$\frac{1}{10}$	1	0	0	0	0	$\frac{27}{10}$
$\frac{3}{5}$	$\frac{2}{5}$	0	-1	1	0	0	6
$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	1	0	6
$\frac{11}{10}$	$\frac{9}{10}$	0	-1	0	0	1	12

$$\text{La s.b.a. è } \begin{vmatrix} x_1 \\ x_2 \\ s_1 \\ \bar{s}_2 \\ t_2 \\ t_3 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ \frac{27}{10} \\ 0 \\ 6 \\ 6 \end{vmatrix} \left. \vphantom{\begin{vmatrix} 0 \\ 0 \\ \frac{27}{10} \\ 0 \\ 6 \\ 6 \end{vmatrix}} \right\} \Psi = 12$$

Nel punto $O \equiv (0, 0)$ Ψ non è minimo

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ		
$\frac{3}{10}$	$\frac{1}{10}$	1	0	0	0	0	$\frac{27}{10}$	$\frac{27}{10} \cdot \frac{10}{3} = 9 \quad \leftarrow \text{min}$
$\frac{3}{5}$	$\frac{2}{5}$	0	-1	1	0	0	6	$6 \cdot \frac{5}{3} = 10$
$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	1	0	6	$6 \cdot 2 = 12$
$\frac{11}{10}$	$\frac{9}{10}$	0	-1	0	0	1	12	

↑

- entra in base x_1 ed esce di base s_1 , prima variabile di base nella base corrente
- nella nuova base le v.n.b. sono x_2 , s_1 ed $\bar{s}_2$ e le v.b. sono, **nell'ordine**, x_1 , t_2 e t_3
- la riga 1 deve esprimere x_1 in funzione delle sole v.n.b., quindi il coefficiente di x_1 deve diventare 1: $\bar{R}_1 = \frac{10}{3} R_1$
- la riga 2 deve esprimere t_2 in funzione delle sole v.n.b., quindi il coefficiente di x_1 deve diventare 0: $\bar{R}_2 = R_2 - \frac{3}{5} \bar{R}_1$
- la riga 3 deve esprimere t_3 in funzione delle sole v.n.b., quindi il coefficiente di x_1 deve diventare 0: $\bar{R}_3 = R_3 - \frac{1}{2} \bar{R}_1$
- la riga 0 deve esprimere ψ in funzione delle sole v.n.b., quindi il coefficiente di x_1 deve diventare 0: $\bar{R}_0 = R_0 - \frac{11}{10} \bar{R}_1$

$$\bar{R}_1 = \frac{10}{3} R_1$$

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
1	$\frac{1}{3}$	$\frac{10}{3}$	0	0	0	0	9

$$\bar{R}_2 = R_2 - \frac{3}{5} \bar{R}_1$$

	x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
R_2	$\frac{3}{5}$	$\frac{2}{5}$	0	-1	1	0	0	6
$-\frac{3}{5} \bar{R}_1$	$-\frac{3}{5}$	$-\frac{1}{5}$	-2	0	0	0	0	$-\frac{27}{5}$
$\bar{R}_2$	0	$\frac{1}{5}$	-2	-1	1	0	0	$\frac{3}{5}$

$$\bar{R}_3 = R_3 - \frac{1}{2} \bar{R}_1$$

	x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
R_3	$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	1	0	6
$-\frac{1}{2} \bar{R}_1$	$-\frac{1}{2}$	$-\frac{1}{6}$	$-\frac{5}{3}$	0	0	0	0	$-\frac{9}{2}$
$\bar{R}_3$	0	$\frac{1}{3}$	$-\frac{5}{3}$	0	0	1	0	$\frac{3}{2}$

$$\bar{R}_0 = R_0 - \frac{11}{10} \bar{R}_1$$

	x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
R_0	$\frac{11}{10}$	$\frac{9}{10}$	0	-1	0	0	1	12
$-\frac{11}{10} \bar{R}_1$	$-\frac{11}{10}$	$-\frac{11}{30}$	$-\frac{11}{3}$	0	0	0	0	$-\frac{99}{10}$
$\bar{R}_0$	0	$\frac{8}{15}$	$-\frac{11}{3}$	-1	0	0	1	$\frac{21}{10}$

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ	
1	$\frac{1}{3}$	$\frac{10}{3}$	0	0	0	0	9
0	$\frac{1}{5}$	-2	-1	1	0	0	$\frac{3}{5}$
0	$\frac{1}{3}$	$-\frac{5}{3}$	0	0	1	0	$\frac{3}{2}$
0	$\frac{8}{15}$	$-\frac{11}{3}$	-1	0	0	1	$\frac{21}{10}$

$$\text{La s.b.a. è } \left[\begin{array}{c} x_1 \\ x_2 \\ s_1 \\ \bar{s}_2 \\ t_2 \\ t_3 \end{array} \right] = \left[\begin{array}{c} 9 \\ 0 \\ 0 \\ 0 \\ \frac{3}{5} \\ \frac{3}{2} \end{array} \right] \left. \vphantom{\begin{array}{c} x_1 \\ x_2 \\ s_1 \\ \bar{s}_2 \\ t_2 \\ t_3 \end{array}} \right\} \Psi = \frac{21}{10}$$

Nel punto $A \equiv (9, 0)$ Ψ non è minimo

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ		
1	$\frac{1}{3}$	$\frac{10}{3}$	0	0	0	0	9	$9 \cdot 3 = 27$
0	$\frac{1}{5}$	-2	-1	1	0	0	$\frac{3}{5}$	$\frac{3}{5} \cdot 5 = 3 \quad \leftarrow \text{min}$
0	$\frac{1}{3}$	$-\frac{5}{3}$	0	0	1	0	$\frac{3}{2}$	$\frac{3}{2} \cdot 3 = \frac{9}{2}$
0	$\frac{8}{15}$	$-\frac{11}{3}$	-1	0	0	1	$\frac{21}{10}$	

$\uparrow$

- entra in base x_2 ed esce di base t_2 , seconda variabile di base nella base corrente
- nella nuova base le v.n.b. sono s_1 , $\bar{s}_2$ e t_2 ; le v.b. sono, **nell'ordine**, x_1 , x_2 e t_3
- la riga 2 deve esprimere x_2 in funzione delle sole v.n.b., quindi il coefficiente di x_2 deve diventare **1**: $\bar{R}_2 = 5 R_2$
- la riga 1 deve esprimere x_1 in funzione delle sole v.n.b., quindi il coefficiente di x_2 deve diventare **0**: $\bar{R}_1 = R_1 - \frac{1}{3} \bar{R}_2$
- la riga 3 deve esprimere t_3 in funzione delle sole v.n.b., quindi il coefficiente di x_2 deve diventare **0**: $\bar{R}_3 = R_3 - \frac{1}{3} \bar{R}_2$
- la riga 0 deve esprimere Ψ in funzione delle sole v.n.b., quindi il coefficiente di x_2 deve diventare **0**: $\bar{R}_0 = R_0 - \frac{8}{15} \bar{R}_2$

$$\bar{R}_2 = 5 R_2 \quad \left| \begin{array}{ccccccc} x_1 & x_2 & s_1 & \bar{s}_2 & t_2 & t_3 & \Psi \\ 0 & 1 & -10 & -5 & 5 & 0 & 0 \end{array} \right. \quad 3$$

$$\bar{R}_1 = R_1 - \frac{1}{3} \bar{R}_2$$

	x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
R_1	1	$\frac{1}{3}$	$\frac{10}{3}$	0	0	0	0	9
$-\frac{1}{3} \bar{R}_2$	0	$-\frac{1}{3}$	$\frac{10}{3}$	$\frac{5}{3}$	$-\frac{5}{3}$	0	0	-1
$\bar{R}_1$	1	0	$\frac{20}{3}$	$\frac{5}{3}$	$-\frac{5}{3}$	0	0	8

$$\bar{R}_3 = R_3 - \frac{1}{3} \bar{R}_2$$

	x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
R_3	0	$\frac{1}{3}$	$-\frac{5}{3}$	0	0	1	0	$\frac{3}{2}$
$-\frac{1}{3} \bar{R}_2$	0	$-\frac{1}{3}$	$\frac{10}{3}$	$\frac{5}{3}$	$-\frac{5}{3}$	0	0	-1
$\bar{R}_3$	0	0	$\frac{5}{3}$	$\frac{5}{3}$	$-\frac{5}{3}$	1	0	$\frac{1}{2}$

$$\bar{R}_0 = R_0 - \frac{8}{15} \bar{R}_2$$

	x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
R_0	0	$\frac{8}{15}$	$-\frac{11}{3}$	-1	0	0	1	$\frac{21}{10}$
$-\frac{8}{15} \bar{R}_2$	0	$-\frac{8}{15}$	$\frac{16}{3}$	$\frac{8}{3}$	$-\frac{8}{3}$	0	0	$-\frac{8}{5}$
$\bar{R}_0$	0	0	$\frac{5}{3}$	$\frac{5}{3}$	$-\frac{8}{3}$	0	1	$\frac{1}{2}$

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
1	0	$\frac{20}{3}$	$\frac{5}{3}$	$-\frac{5}{3}$	0	0	8
0	1	-10	-5	5	0	0	3
0	0	$\frac{5}{3}$	$\frac{5}{3}$	$-\frac{5}{3}$	1	0	$\frac{1}{2}$
0	0	$\frac{5}{3}$	$\frac{5}{3}$	$-\frac{8}{3}$	0	1	$\frac{1}{2}$

La s.b.a. è

$$\begin{array}{c|c} \begin{array}{c} x_1 \\ x_2 \\ s_1 \\ \bar{s}_2 \\ t_2 \\ t_3 \end{array} & \begin{array}{c} 8 \\ 3 \\ 0 \\ 0 \\ 0 \\ \frac{1}{2} \end{array} \end{array} \quad \left. \vphantom{\begin{array}{c} x_1 \\ x_2 \\ s_1 \\ \bar{s}_2 \\ t_2 \\ t_3 \end{array}} \right\} \psi = \frac{1}{2}$$

Nel punto $B \equiv (8, 3)$ ψ non è minimo

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ		
1	0	$\frac{20}{3}$	$\frac{5}{3}$	$-\frac{5}{3}$	0	0	8	$8 \cdot \frac{3}{20} = \frac{12}{10}$
0	1	-10	-5	5	0	0	3	
0	0	$\frac{5}{3}$	$\frac{5}{3}$	$-\frac{5}{3}$	1	0	$\frac{1}{2}$	$\frac{1}{2} \cdot \frac{3}{5} = \frac{3}{10} \leftarrow \text{min}$
0	0	$\frac{5}{3}$	$\frac{5}{3}$	$-\frac{8}{3}$	0	1	$\frac{1}{2}$	

$\uparrow$

- entra in base s_1 ed esce di base t_3 , terza variabile di base nella base corrente
- nella nuova base le v.n.b. sono $\bar{s}_2$, t_2 e t_3 ; le v.b. sono, **nell'ordine**, x_1 , x_2 e s_1
- la riga 3 deve esprimere s_1 in funzione delle sole v.n.b., quindi il coefficiente di s_1 deve diventare **1**: $\bar{R}_3 = \frac{3}{5} R_3$
- la riga 1 deve esprimere x_1 in funzione delle sole v.n.b., quindi il coefficiente di s_1 deve diventare **0**: $\bar{R}_1 = R_1 - \frac{20}{3} \bar{R}_3$
- la riga 2 deve esprimere x_2 in funzione delle sole v.n.b., quindi il coefficiente di s_1 deve diventare **0**: $\bar{R}_2 = R_2 + 10 \bar{R}_3$
- la riga 0 deve esprimere Ψ in funzione delle sole v.n.b., quindi il coefficiente di x_2 deve diventare **0**: $\bar{R}_0 = R_0 - \frac{5}{3} \bar{R}_3$

$$\bar{R}_3 = \frac{3}{5} R_3 \quad \begin{array}{c|ccccccc} & x_1 & x_2 & s_1 & \bar{s}_2 & t_2 & t_3 & \Psi \\ \hline & 0 & 0 & 1 & 1 & -1 & \frac{3}{5} & 0 \end{array}$$

$$\bar{R}_1 = R_1 - \frac{20}{3} \bar{R}_3$$

	x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
R_1	1	0	$\frac{20}{3}$	$\frac{5}{3}$	$-\frac{5}{3}$	0	0	8
$-\frac{20}{3} \bar{R}_3$	0	0	$-\frac{20}{3}$	$-\frac{20}{3}$	$\frac{20}{3}$	-4	0	-2
$\bar{R}_1$	1	0	0	-5	5	-4	0	6

$$\bar{R}_2 = R_2 + 10 \bar{R}_3$$

	x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
R_2	0	1	-10	-5	5	0	0	3
$+10 \bar{R}_3$	0	0	10	10	-10	6	0	3
$\bar{R}_2$	0	1	0	5	-5	6	0	6

$$\bar{R}_0 = R_0 - \frac{5}{3} \bar{R}_3$$

	x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
R_0	0	0	$\frac{5}{3}$	$\frac{5}{3}$	$-\frac{8}{3}$	0	1	$\frac{1}{2}$
$-\frac{5}{3} \bar{R}_3$	0	0	$-\frac{5}{3}$	$-\frac{5}{3}$	$\frac{5}{3}$	-1	0	$-\frac{1}{2}$
$\bar{R}_0$	0	0	0	0	-1	-1	1	0

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ	
1	0	0	-5	5	-4	0	6
0	1	0	5	-5	6	0	6
0	0	1	1	-1	$\frac{3}{5}$	0	$\frac{3}{10}$
0	0	0	0	-1	-1	1	0

$$\text{La s.b.a. è } \begin{array}{c} x_1 \\ x_2 \\ s_1 \\ \bar{s}_2 \\ t_2 \\ t_3 \end{array} = \begin{array}{c} 6 \\ 6 \\ \frac{3}{10} \\ 0 \\ 0 \\ 0 \end{array} \left. \vphantom{\begin{array}{c} x_1 \\ x_2 \\ s_1 \\ \bar{s}_2 \\ t_2 \\ t_3 \end{array}} \right\} \Psi = 0$$

Nel punto $D \equiv (6, 6)$ Ψ è minimo

La soluzione di base $x_1 = x_2 = 6$ è **ammissibile per il problema originario** perché $\Psi = 0$.

In alternativa:

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
1	0	5	0	0	-1	0	$\frac{15}{2}$
0	1	-5	0	0	3	0	$\frac{9}{2}$
0	0	1	1	-1	$\frac{3}{5}$	0	$\frac{3}{10}$
0	0	0	0	-1	-1	1	0

$$\text{La s.b.a. è } \begin{array}{c} \left| \begin{array}{c} x_1 \\ x_2 \\ s_1 \\ \bar{s}_2 \\ t_2 \\ t_3 \end{array} \right| = \left| \begin{array}{c} \frac{15}{2} \\ \frac{9}{2} \\ 0 \\ \frac{3}{10} \\ 0 \\ 0 \end{array} \right| \end{array} \left. \vphantom{\begin{array}{c} \left| \begin{array}{c} x_1 \\ x_2 \\ s_1 \\ \bar{s}_2 \\ t_2 \\ t_3 \end{array} \right|} \right\} \psi = 0$$

punto $C \equiv \left(\frac{15}{2}, \frac{9}{2}\right)$: **ottimo**

FASE 2 : ricerca di una s.b.a. ottima

- le variabili t_2 e t_3 possono essere eliminate dal tableau,
perché nel problema originario devono necessariamente assumere valore nullo
(se invece le si mantiene, si ottengono tutte le colonne dell'inversa della base)

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	ψ	
1	0	0	-5	5	-4	0	6
0	1	0	5	-5	6	0	6
0	0	1	1	-1	$\frac{3}{5}$	0	$\frac{3}{10}$
0	0	0	0	-1	-1	1	0

- viene considerata la f.o. originaria, che però non è in forma canonica perché la variabile obiettivo φ non è espressa in funzione delle sole v.n.b.

x_1	x_2	s_1	$\bar{s}_2$	φ	
1	0	0	-5	0	6
0	1	0	5	0	6
0	0	1	1	0	$\frac{3}{10}$
$-\frac{2}{5}$	$-\frac{1}{2}$	0	0	1	0

$$\bar{R}_0 = R_0 + \frac{2}{5} R_1 + \frac{1}{2} R_2$$

	x_1	x_2	s_1	$\bar{s}_2$	φ	
R_0	$-\frac{2}{5}$	$-\frac{1}{2}$	0	0	1	0
$+\frac{2}{5}R_1$	$\frac{2}{5}$	0	0	-2	0	$\frac{12}{5}$
$+\frac{1}{2}R_2$	0	$\frac{1}{2}$	0	$\frac{5}{2}$	0	3
$= \bar{R}_0$	0	0	0	$\frac{1}{2}$	1	$\frac{27}{5}$

x_1	x_2	s_1	$\bar{s}_2$	φ	
1	0	0	-5	0	6
0	1	0	5	0	6
0	0	1	1	0	$\frac{3}{10}$
0	0	0	$\frac{1}{2}$	1	$\frac{27}{5}$

l'equazione che definisce la variabile obiettivo
è in forma canonica

Punto D : s.b.a. non ottima per il problema originario

x_1	x_2	s_1	$\bar{s}_2$	φ	
1	0	5	0	0	$\frac{15}{2}$
0	1	-5	0	0	$\frac{9}{2}$
0	0	1	1	0	$\frac{3}{10}$
0	0	$-\frac{1}{2}$	0	1	$\frac{21}{4}$

Punto C : s.b.a. ottima per il problema originario

Problema con regione ammissibile vuota

$$\min \varphi = 0.4 x_1 + 0.5 x_2$$

minimize the radiation absorbed by healthy tissues

$$\text{s.t. } 0.3 x_1 + 0.1 x_2 \leq 2.7$$

upper bound to radiation absorbed by critical tissues

$$0.6 x_1 + 0.4 x_2 \geq 6$$

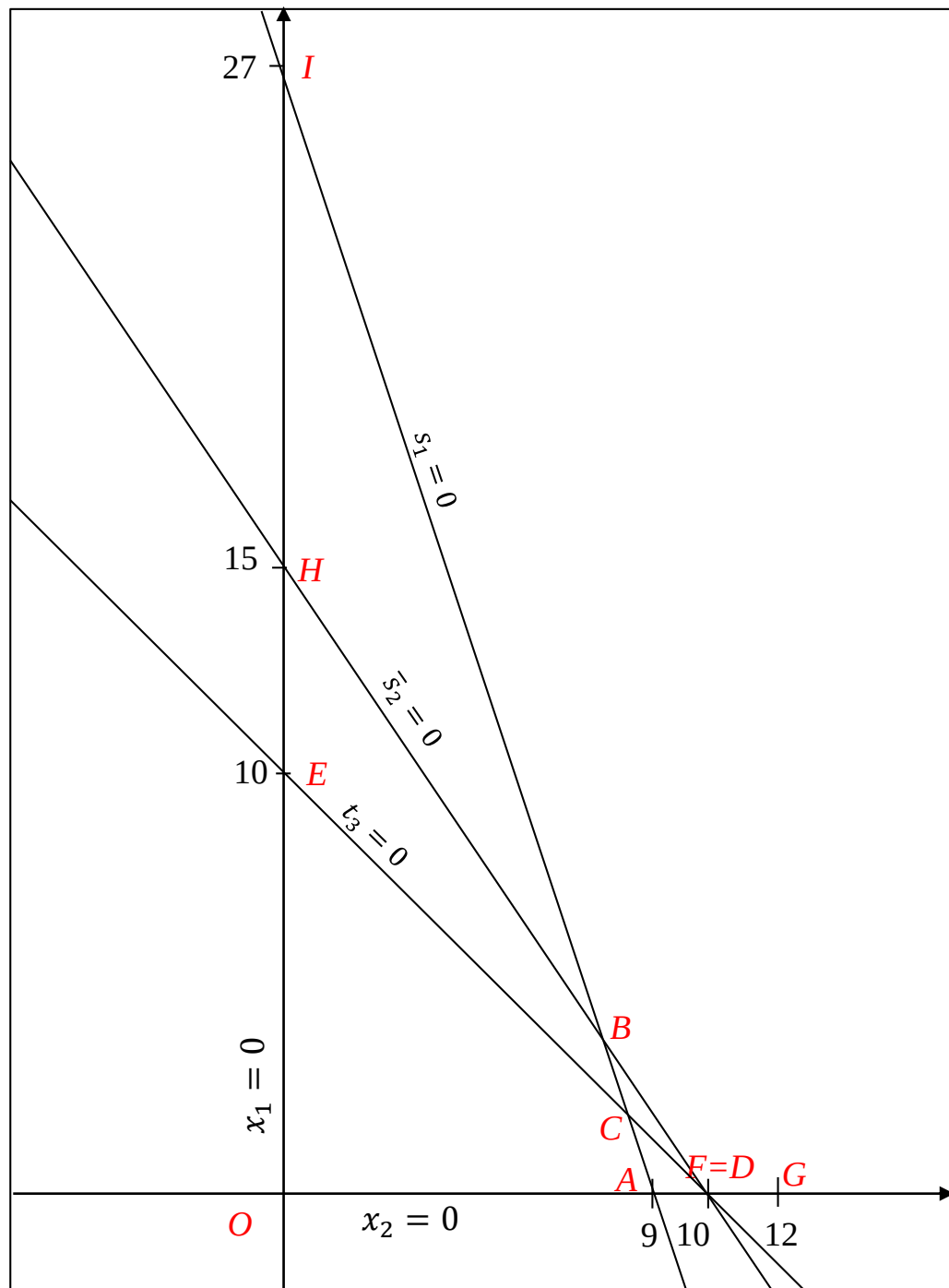
lower bound to the radiation absorbed by the tumor core

$$0.5 x_1 + 0.5 x_2 = 5$$

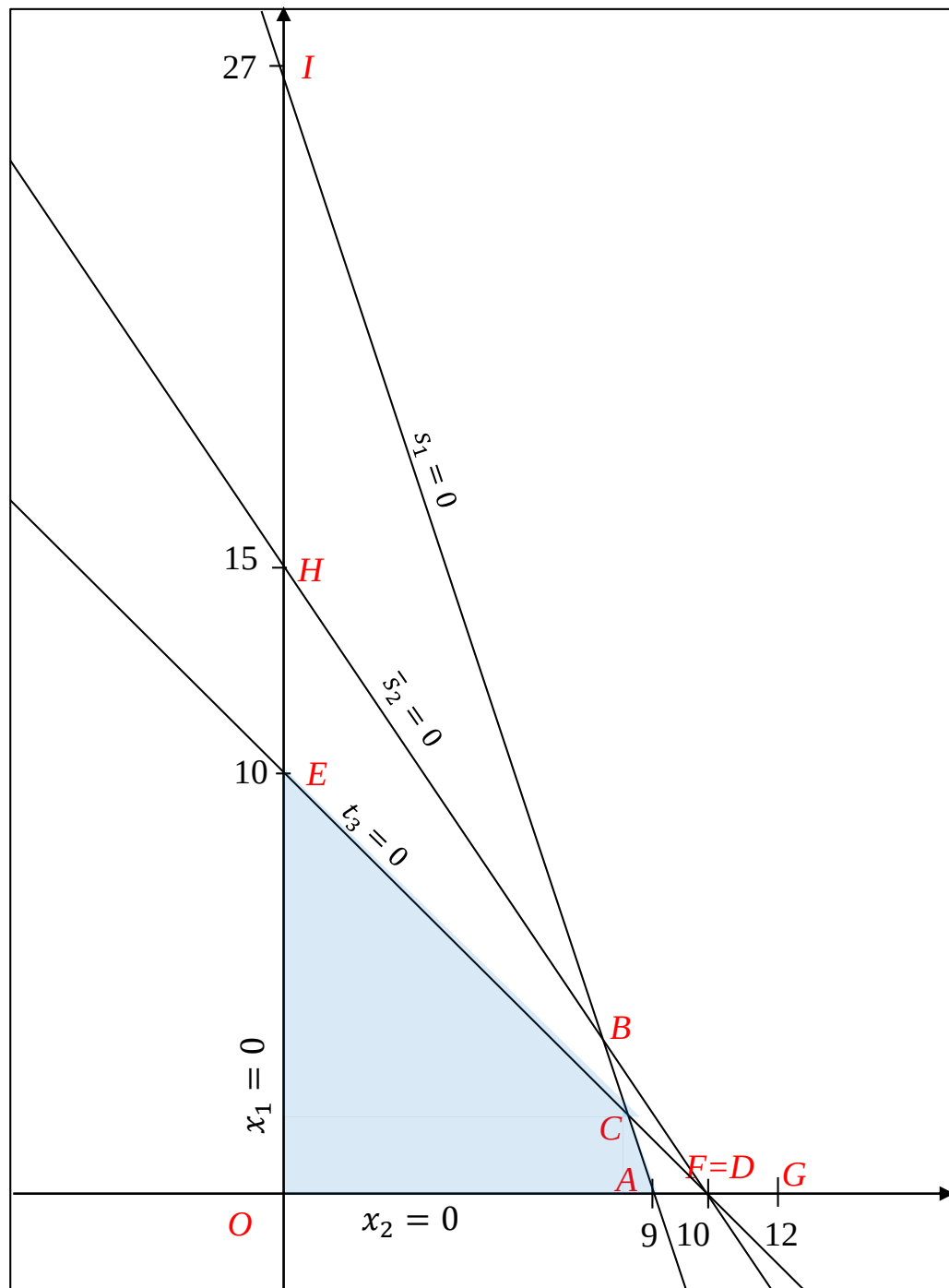
absorbed radiation required in the tumor region

$$x_1 \geq 0, x_2 \geq 0$$

the intensities of rays 1 and 2 at their respective entry points are expressed by nonnegative real numbers



The feasible region
of the original problem is empty.



$$0.3 x_1 + 0.1 x_2 + s_1 = 2.7$$

$$0.6 x_1 + 0.4 x_2 - \bar{s}_2 + t_2 = 6$$

$$0.5 x_1 + 0.5 x_2 + t_3 = 5$$

$$x_1 \geq 0, x_2 \geq 0, s_1 \geq 0, \bar{s}_2 \geq 0, t_2 > 0, t_3 > 0$$

The feasible region of the auxiliary problem is the polytope of vertices O, A, C, E .

	x_1	x_2	t_2	t_3	$\Psi = t_2 + t_3$
O	0	0	6	5	11
A	9	0	$\frac{3}{5}$	$\frac{1}{2}$	$\frac{11}{10}$
C	$\frac{17}{2}$	$\frac{3}{2}$	$\frac{3}{10}$	0	$\frac{3}{10}$
E	0	10	2	0	2

where

$$t_2 = \max \left\{ 0, \left(6 - \frac{3}{5} x_1 - \frac{2}{5} x_2 \right) \right\}$$

$$t_3 = \max \left\{ 0, \left(5 - \frac{1}{2} x_1 - \frac{1}{2} x_2 \right) \right\}$$

Problema ausiliario: il tableau associato al vertice $(x_1, x_2) = (0, 0)$ è

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ	
$\frac{3}{10}$	$\frac{1}{10}$	1	0	0	0	0	$\frac{27}{10}$
$\frac{3}{5}$	$\frac{2}{5}$	0	-1	1	0	0	6
$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	1	0	5
$\frac{11}{10}$	$\frac{9}{10}$	0	-1	0	0	1	11

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ	
$\frac{3}{10}$	$\frac{1}{10}$	1	0	0	0	0	$\frac{27}{10}$
$\frac{3}{5}$	$\frac{2}{5}$	0	-1	1	0	0	6
$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	1	0	5
$\frac{11}{10}$	$\frac{9}{10}$	0	-1	0	0	1	11

$\frac{27}{10} \cdot \frac{10}{3} = 9$
 $6 \cdot \frac{5}{3} = 10$
 $5 \cdot 2 = 10$

$\leftarrow \text{min}$

$\uparrow$

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ		
1	$\frac{1}{3}$	$\frac{10}{3}$	0	0	0	0	9	$9 \cdot 3 = 27$
0	$\frac{1}{5}$	-2	-1	1	0	0	$\frac{3}{5}$	$\frac{3}{5} \cdot 5 = 3$
0	$\frac{1}{3}$	$-\frac{5}{3}$	0	0	1	0	$\frac{1}{2}$	$\frac{1}{2} \cdot 3 = \frac{3}{2} \leftarrow \text{min}$
0	$\frac{8}{15}$	$-\frac{11}{3}$	-1	0	0	1	$\frac{11}{10}$	

$\uparrow$

x_1	x_2	s_1	$\bar{s}_2$	t_2	t_3	Ψ	
1	0	$\frac{5}{3}$	0	0	-1	0	$\frac{17}{2}$
0	0	-1	-1	1	$-\frac{3}{5}$	0	$\frac{3}{10}$
0	1	-5	0	0	3	0	$\frac{3}{2}$
0	0	-1	-1	0	$-\frac{8}{5}$	1	$\frac{3}{10}$

punto $C \equiv \left(\frac{17}{2}, \frac{3}{2}\right)$: **punto di minimo del problema ausiliario**

$\Psi = \frac{3}{10}$ è il valore minimo della funzione Ψ sull'insieme delle s.b.a. del problema ausiliario

$\Rightarrow$ non esiste nessuna s.b.a. con valore di inammissibilità $\Psi = 0$

Teorema Fondamentale della PL (1ª parte): se esiste un s.a., allora esiste un s.b.a.

$\Rightarrow$ se non esistono s.b.a., allora non esistono s.a.

$\Rightarrow$ la regione ammissibile del problema originario è vuota

Appendix

The proof of the **Fundamental Theorem of Linear Programming**

Given the LP problem

$$\begin{aligned} \min \varphi = & \quad \mathbf{c}_w^T \mathbf{w} + \gamma \\ \text{s.t.} \quad & \quad \mathbf{F} \mathbf{w} = \mathbf{b} \quad \mathbf{F} \in \mathbb{R}^{m, n+m}, \text{rank}(\mathbf{F}) = m \\ & \quad \mathbf{w} \geq 0 \end{aligned}$$

- 1) if feasible solutions exist, then at least one of them is basic.
- 2) if optimal feasible solutions exist, then at least one of them is basic.

Let $\mathbf{w}$ be a *feasible solution* of $\mathbf{F} \mathbf{w} = \mathbf{b}$ (*)

Let p denote the number of strictly positive components of $\mathbf{w}$ and

let the **strictly positive** components of $\mathbf{w}$ be **the first p components** (**)

$$\begin{vmatrix} f_{1,1} & f_{1,2} & \vdots & f_{1,p} & f_{1,p+1} & \vdots & \vdots & f_{1,n+m} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ f_{m,1} & f_{m,2} & \vdots & f_{m,p} & f_{m,p+1} & \vdots & \vdots & f_{m,n+m} \end{vmatrix} \begin{vmatrix} w_1 \\ w_2 \\ \vdots \\ w_p \\ 0 \\ \vdots \\ \vdots \\ 0 \end{vmatrix} = \begin{vmatrix} b_1 \\ \vdots \\ \vdots \\ \vdots \\ b_m \end{vmatrix}$$

(*) recall that $\mathbf{w}$ is called a **solution** if it satisfies $\mathbf{F} \mathbf{w} = \mathbf{b}$ and it is **feasible** if $\mathbf{w} \geq \mathbf{0}$

(**) if the strictly positive components of $\mathbf{w}$ are not the first p , we reorder the components of $\mathbf{w}$ so as to bring the strictly positive ones in the first p position

and the columns of $\mathbf{F}$ and the components of $\mathbf{c}_w = \begin{bmatrix} \mathbf{c} \\ \mathbf{0} \end{bmatrix}$ correspondingly

$$\left| \begin{array}{cccc|cccc}
 f_{1,1} & f_{1,2} & \vdots & f_{1,p} & f_{1,p+1} & \vdots & \vdots & f_{1,n+m} \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 f_{m,1} & f_{m,2} & \vdots & f_{m,p} & f_{m,p+1} & \vdots & \vdots & f_{m,n+m}
 \end{array} \right| \begin{array}{c} w_1 \\ w_2 \\ \vdots \\ w_p \\ \hline 0 \\ \vdots \\ \vdots \\ 0 \end{array} = \begin{array}{c} b_1 \\ \vdots \\ \vdots \\ \vdots \\ b_m \end{array}$$

By defining $\mathbf{F} = [\mathbf{F}_p \quad \mathbf{F}_{n+m-p}]$, $\mathbf{c}_w = \begin{bmatrix} \mathbf{c}_p \\ \mathbf{c}_{n+m-p} \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} \mathbf{w}_p \\ \mathbf{0} \end{bmatrix}$, with $\mathbf{w}_p > \mathbf{0}$ by hypothesis,

we can express the above linear system in the more compact form $\mathbf{F}_p \mathbf{w}_p + \mathbf{F}_{n+m-p} \mathbf{0} = \mathbf{b}$

and the objective function as $\mathbf{c}_p^T \mathbf{w}_p + \gamma$

$$\begin{vmatrix} f_{1,1} & f_{1,2} & \vdots & f_{1,p} \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ f_{m,1} & f_{m,2} & \vdots & f_{m,p} \end{vmatrix} \begin{vmatrix} w_1 \\ w_2 \\ \vdots \\ w_p \end{vmatrix} = \begin{vmatrix} b_1 \\ \vdots \\ \vdots \\ \vdots \\ b_m \end{vmatrix}$$

Note now that two cases may occur:

- 1) either the columns of $\mathbf{F}_p$ are linearly independent ($\mathbf{F}_p \mathbf{v} = \mathbf{0} \leftrightarrow \mathbf{v} = \mathbf{0}, \mathbf{v} \in \mathbb{R}^p$)
- 2) or they are linearly dependent ($\exists \mathbf{v} \neq \mathbf{0} : \mathbf{F}_p \mathbf{v} = \mathbf{0}, \mathbf{v} \in \mathbb{R}^p$)

Case 1: the columns of $\mathbf{F}_p$ are linearly independent

In this case, $\mathbf{w}$ is a **basic feasible solution**, either **non degenerate** or **degenerate**

➤ **non degenerate**, if $p = m$:

- $\mathbf{F}_p$ is the associated basis (it is an $m \times m$ nonsingular matrix)
- all basic variables are strictly positive

➤ **degenerate**, if $p < m$:

- the associated basis consists of the $p < m$ columns of $\mathbf{F}_p$ and of $m - p$ columns selected from $\mathbf{F}_{n+m-p}$ and linearly independent on the columns of $\mathbf{F}_p$ (such selection is always possible, being $\text{rank}(\mathbf{F}) = m$)
- $m - p$ basic variables are zero

Case 2: the columns of $\mathbf{F}_p$ are linearly dependent

In this case, we can construct a new **feasible solution** $\tilde{\mathbf{w}}$ having no more than $p - 1$ **positive components** (at least one more zero component than the initial feasible solution $\mathbf{w}$, i.e. the point is at the intersection of at least one more hyperplane)

The new **feasible solution** $\tilde{\mathbf{w}}$ is obtained from $\mathbf{w}$ by a suitably chosen vector $\mathbf{d}$ multiplied by a scalar $\alpha \neq 0$

$$\begin{array}{c} \left| \begin{array}{c} w_1 \\ w_2 \\ \vdots \\ w_p \\ \hline 0 \\ \vdots \\ \vdots \\ 0 \end{array} \right| + \left| \begin{array}{c} d_1 \\ d_2 \\ \vdots \\ d_p \\ \hline 0 \\ \vdots \\ \vdots \\ 0 \end{array} \right| \alpha = \left| \begin{array}{c} w_1 + d_1 \alpha \\ w_2 + d_2 \alpha \\ \vdots \\ w_p + d_p \alpha \\ \hline 0 \\ \vdots \\ \vdots \\ 0 \end{array} \right|$$

$$\begin{bmatrix} w_p \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} d_p \\ \mathbf{0} \end{bmatrix} \alpha = \begin{bmatrix} w_p + d_p \alpha \\ \mathbf{0} \end{bmatrix}$$

$\tilde{\mathbf{w}}$ must be a feasible solution, i.e.

$$\mathbf{F} \tilde{\mathbf{w}} = \mathbf{b}, \tilde{\mathbf{w}} \geq \mathbf{0}$$

$$\left| \begin{array}{cccc|cccc} f_{1,1} & f_{1,2} & \vdots & f_{1,p} & f_{1,p+1} & \vdots & \vdots & f_{1,n+m} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ f_{m,1} & f_{m,2} & \vdots & f_{m,p} & f_{m,p+1} & \vdots & \vdots & f_{m,n+m} \end{array} \right| \left| \begin{array}{c} w_1 + d_1 \alpha \\ w_2 + d_2 \alpha \\ \vdots \\ w_p + d_p \alpha \\ \hline 0 \\ \vdots \\ \vdots \\ 0 \end{array} \right| = \left| \begin{array}{c} b_1 \\ \vdots \\ \vdots \\ \vdots \\ b_m \end{array} \right|$$

$$[\mathbf{F}_p \quad \mathbf{F}_{n+m-p}] \begin{bmatrix} \mathbf{w}_p + \mathbf{d}_p \alpha \\ \mathbf{0} \end{bmatrix} = \mathbf{b}$$

$$\mathbf{F}_p (\mathbf{w}_p + \mathbf{d}_p \alpha) + \mathbf{F}_{n+m-p} \mathbf{0} = \mathbf{b}$$

$$\underbrace{\mathbf{F}_p \mathbf{w}_p}_{\mathbf{b}} + \mathbf{F}_p \mathbf{d}_p \alpha = \mathbf{b}$$

$$\Rightarrow \mathbf{F}_p \mathbf{d}_p \alpha = \mathbf{0}$$

$$\Rightarrow \text{being } \alpha \neq 0, \text{ then } \mathbf{F}_p \mathbf{d}_p = \mathbf{0}$$

Since in this case the columns of $\mathbf{F}_p$ are linearly dependent, then $\exists \mathbf{d}_p \neq \mathbf{0} : \mathbf{F}_p \mathbf{d}_p = \mathbf{0}$,

i.e. at least one of its components is nonzero

Conditions on the step length α

The nonzero value of α is determined by the nonnegativity conditions

$$\tilde{w}_p = w_p + d_p \alpha \geq 0$$

For each k -th component, $1 \leq k \leq p$, one out of the three following cases occurs:

- $w_k > 0$ and $d_k = 0$:

$$\tilde{w}_k = w_k > 0 \text{ for all } \alpha \in \mathbb{R}$$

- $w_k > 0$ and $d_k > 0$:

$$\tilde{w}_k = w_k + d_k \alpha \geq 0 \quad \text{for } \alpha \geq -\frac{w_k}{d_k}$$



- $w_k > 0$ and $d_k < 0$:

$$\tilde{w}_k = w_k + d_k \alpha \geq 0 \quad \text{for } \alpha \leq -\frac{w_k}{d_k}$$



Let k^* ($1 \leq k^* \leq p$) be an index for which the absolute value of $-\frac{w_{k^*}}{d_{k^*}}$ is minimum.

By assigning α the (nonzero) value $\alpha^* = -\frac{w_{k^*}}{d_{k^*}}$ it follows that

- the k^* -component of $\tilde{w}_p$ is zero
- all other components of $\tilde{w}_p$ are nonnegative.

Note: more than one component of $\tilde{w}_p$ is zero if $-\frac{w_k}{d_k} = \alpha^*$ for some $k \neq k^*$

The solution $\tilde{w}$ we have obtained is a feasible solution with $\tilde{p} \leq p - 1$.

Now

- either $\mathbf{F}_{\tilde{p}}$ is full rank and $\tilde{\mathbf{w}}$ is a basic feasible solution
- or $\mathbf{F}_{\tilde{p}}$ has linearly dependent columns, in which case a new feasible solution $\tilde{\tilde{\mathbf{w}}}$ can be constructed, with $\tilde{\tilde{p}} \leq \tilde{p} - 1$

By iterating this procedure a finite number of times, a basic feasible solution $\hat{\mathbf{w}}$ is obtained

(i.e. the corresponding $\mathbf{F}_{\hat{p}}$ has linearly independent columns)

Q. E. D. (part 1)

Note :

the solution $\tilde{\mathbf{w}} = \mathbf{w} + \mathbf{d} \alpha$ is feasible for all values of α such that

$$-\left| \frac{w_{k^*}}{d_{k^*}} \right| \leq \alpha \leq \left| \frac{w_{k^*}}{d_{k^*}} \right|$$

This observation will be useful in the proof of Part 2.

Proof of Part 2

Objective function;

$$\varphi = \mathbf{c}_p^T (\mathbf{w}_p + \mathbf{d}_p \alpha) + \mathbf{c}_{n+m-p}^T + \gamma$$

$$\varphi = \mathbf{c}_p^T \mathbf{w}_p + \mathbf{c}_p^T \mathbf{d}_p \alpha + \gamma$$

Let $\mathbf{w}^*$ be an optimal feasible solution, i.e.

$$\underbrace{\mathbf{c}_w^{*T} \mathbf{w} + \gamma}_{\varphi(\mathbf{w}^*)} \leq \underbrace{\mathbf{c}_w^T \mathbf{w} + \gamma}_{\varphi(\mathbf{w})}$$

for all feasible solutions $\mathbf{w}$, i.e. vectors $\mathbf{w}$ such that $\mathbf{F} \mathbf{w} = \mathbf{b}$ and $\mathbf{w} \geq \mathbf{0}$

If $\mathbf{w}^*$ is a non basic solution, then, by the procedure used in the proof of Part 1,

a sequence of optimal feasible solutions is constructed

until an optimal basic feasible solution is determined.

The objective function value in the initial optimal feasible solution $\mathbf{w}^*$ is

$$\varphi^* = \mathbf{c}_p^T \mathbf{w}_p^* + \gamma$$

Recall that after the proof of Part 1 we noted that

the solution $\tilde{\mathbf{w}}^*(\alpha) = \mathbf{w}^* + \mathbf{d} \alpha$ is feasible for all values of α such that

$$-\alpha^* \leq \alpha \leq \alpha^*, \text{ with } \alpha^* = \left| \frac{w_{k^*}}{d_{k^*}} \right|.$$

In these points the objective function value is $\tilde{\varphi}^*(\alpha) = \underbrace{\mathbf{c}_p^T \mathbf{w}_p^*}_{\varphi^*} + \mathbf{c}_p^T \mathbf{d}_p \alpha$

If $\mathbf{c}_p^T \mathbf{d}_p > 0$, then $\tilde{\varphi}^*(\alpha) < \varphi^* \quad -|\alpha^*| \leq \alpha \leq 0$

(i.e. in the left interval some points have lower objective values than $\mathbf{w}^*$)

If $\mathbf{c}_p^T \mathbf{d}_p < 0$, then $\tilde{\varphi}^*(\alpha) < \varphi^* \quad 0 \leq \alpha \leq |\alpha^*|$

(i.e. in the right interval some points have lower objective values than $\mathbf{w}^*$)

Since $\mathbf{c}_p^T \mathbf{d}_p > 0$ and $\mathbf{c}_p^T \mathbf{d}_p < 0$ contradict the assumption that $\mathbf{w}$ is optimal, then $\mathbf{c}_p^T \mathbf{d}_p = 0$,

which implies that all elements constructed by the procedure are optimal.

Q. E. D.

Consequences of the fundamental theorem of Linear Programming

(1)

If there are no basic feasible solutions, there are no feasible solutions
(i.e. the feasible region is empty)

This result is used in the «feasibility phase» of the solution algorithm, in which

- either a basic feasible solution is determined
- or it is shown that no basic feasible solutions exist,
which implies that the feasible region is empty (infeasible LP problem)

Consequences of the fundamental theorem of Linear Programming

(2)

If the feasible region is non empty and the objective function has a minimum, then at least one of its minimizers is a basic feasible solution.

This result is used in the «optimality phase» of the solution algorithm, which consists in generating a sequence of basic feasible solutions until

- either an optimal basic feasible solution is found
- or the problem is shown to be unbounded.

The optimal solution is sought in the subset of the basic feasible solutions which has finite cardinality
(while in general the feasible region has infinitely many solutions).